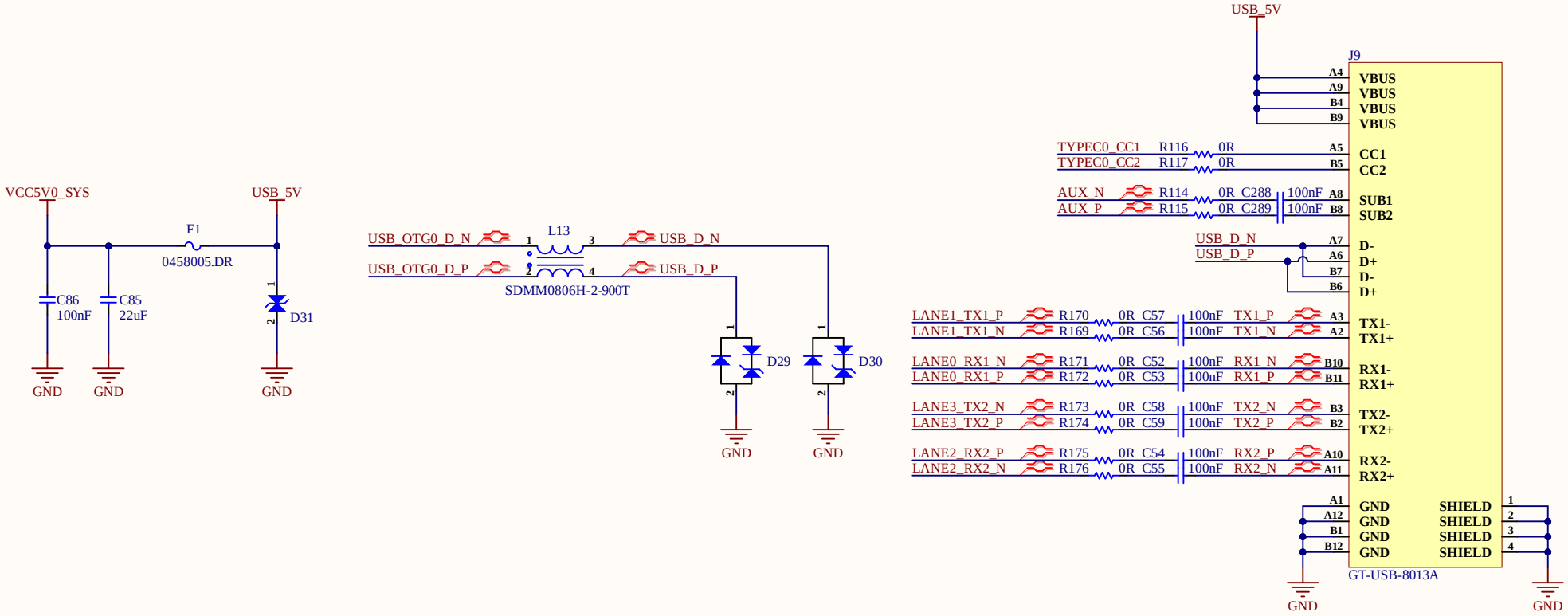
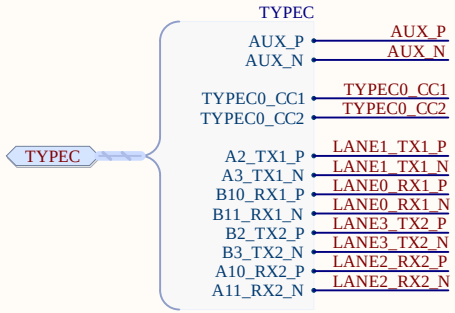
TYPE-C_INTERFACE_SHEET





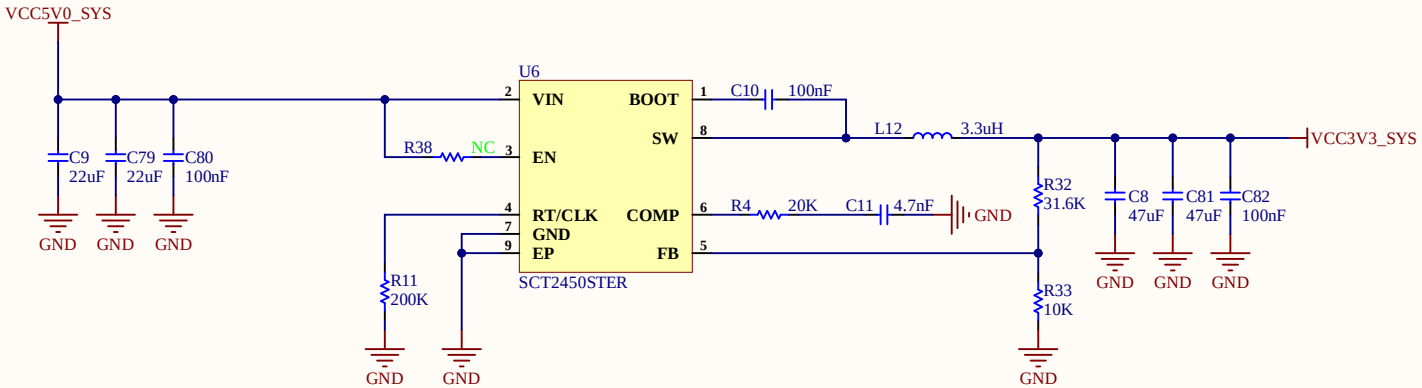




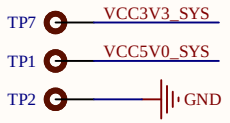
eSpark Consulting Group

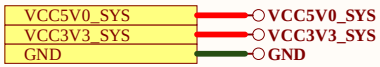
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File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 26/11/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 23

POWER_LDO_SHEET



TEST POINTs

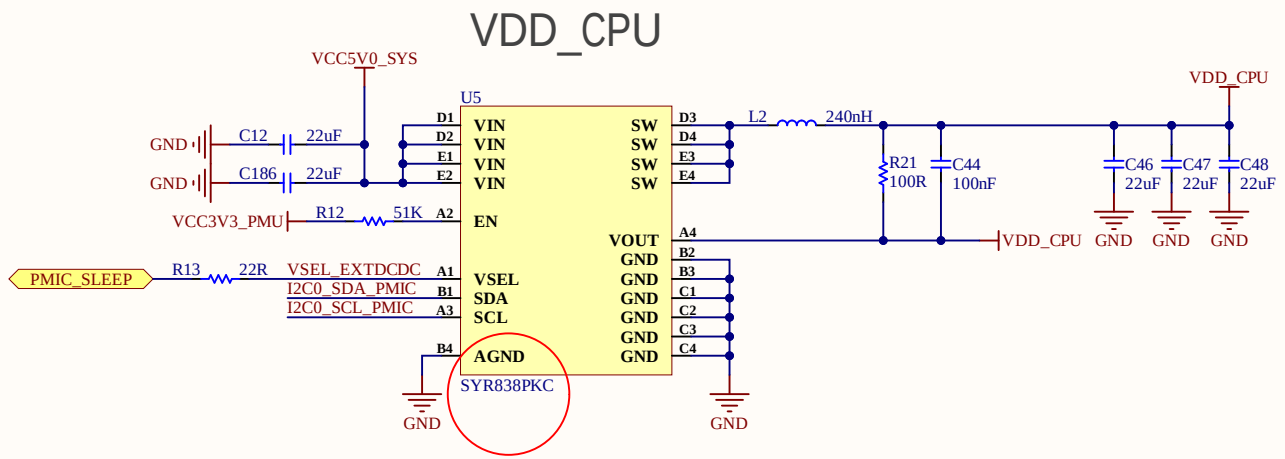


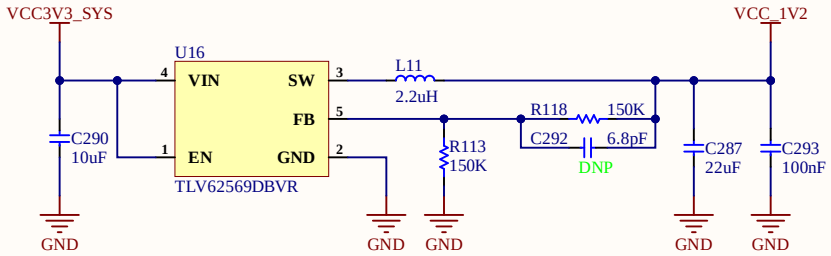


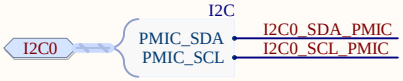
eSpark Consulting Group

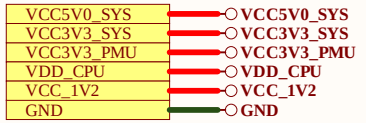
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File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 21/11/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 23

POWER_OTHER_SHEET





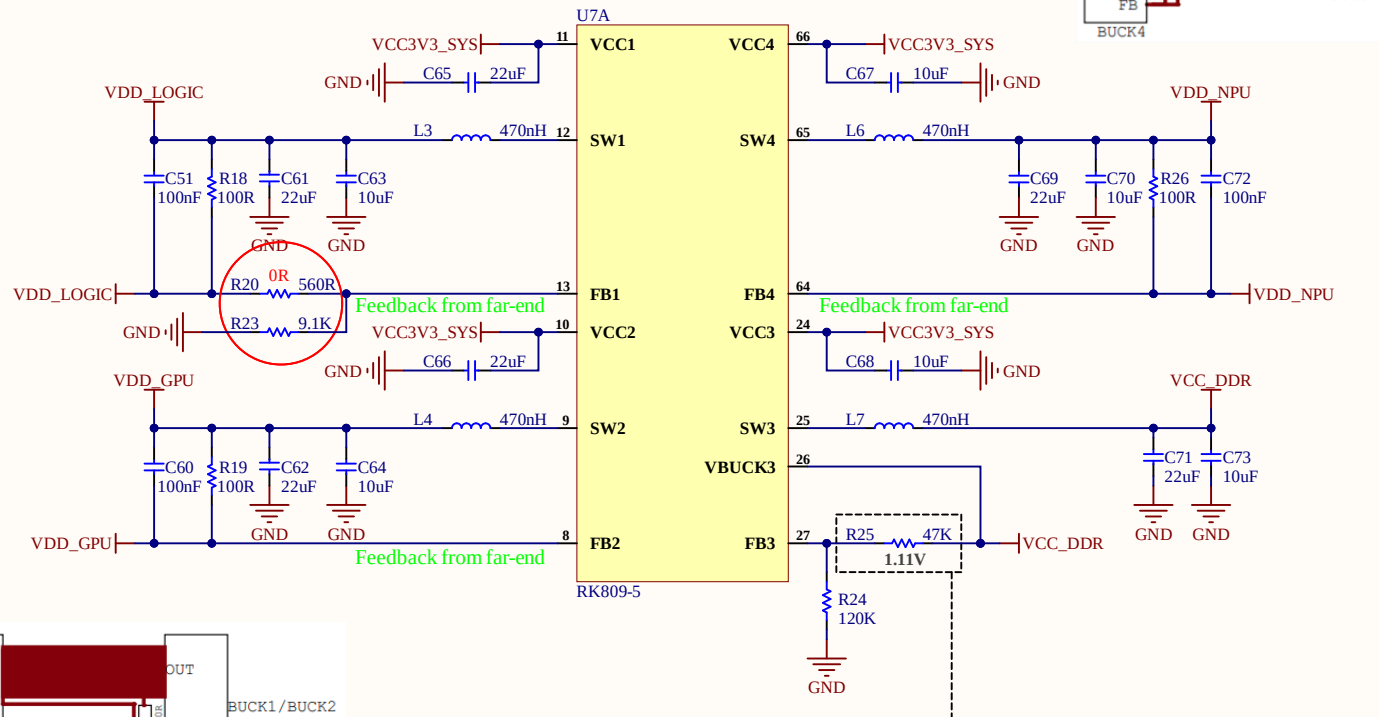




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Title: POWER_OTHER_SHEET.SchDoc	
File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 28/10/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 23

PMIC RK809 DC-DC



DDR4	1.21V	62K 1%
LPDDR4/4x	1.11V	47K 1%
DDR3	1.53V	110K 1%
DDR3L	1.35V	82K 1%
LPDDR3	1.25V	68K 1%

VCC3V3_SYS	VCC3V3_SYS
VDD_LOGIC	VDD_LOGIC
VDD_GPU	VDD_GPU
VDD_NPU	VDD_NPU
VCC_DDR	VCC_DDR
GND	GND

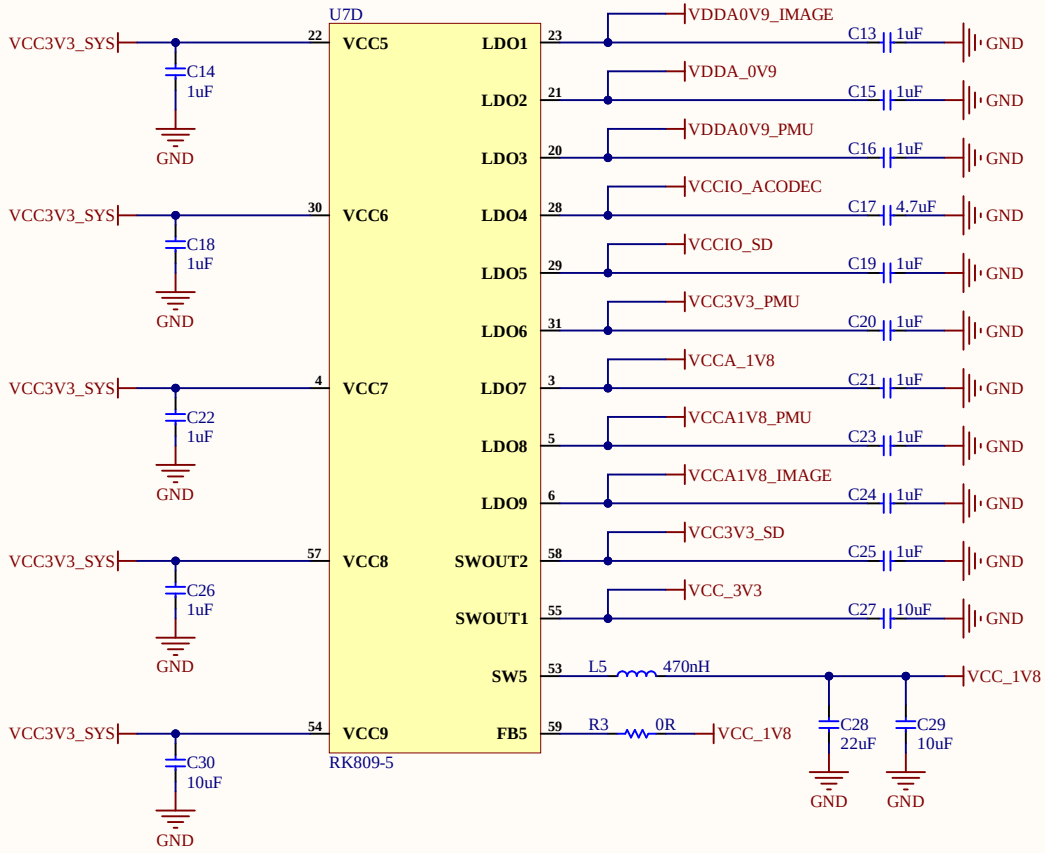
eSpark Consulting Group

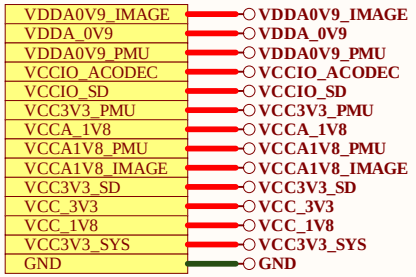
Title: POWER_PMIC_RK809_DC DC.SchDoc

File: TYPEC_4KTransmitter-v1.0.PrjPcb

Date: 25/10/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 23

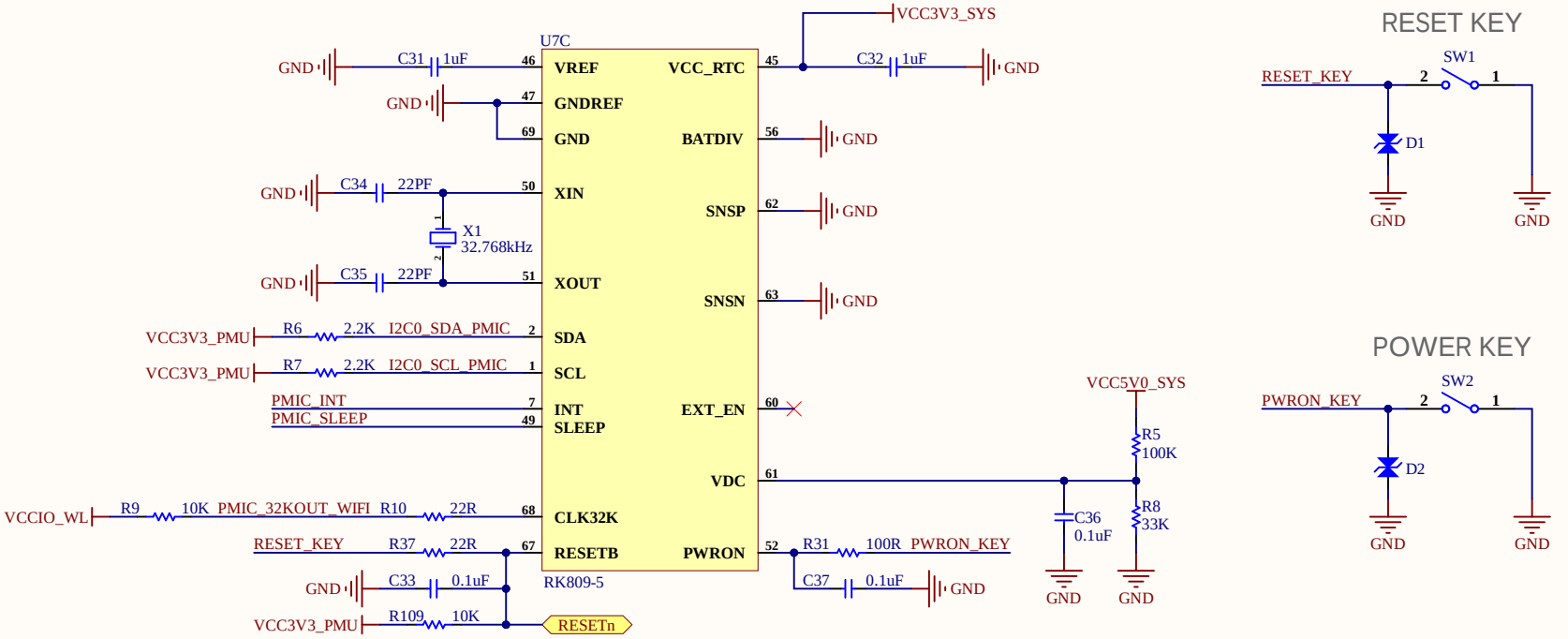
PMIC RK809 LDO

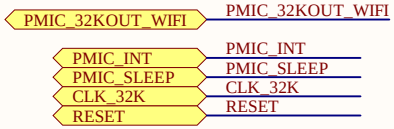


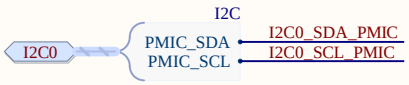


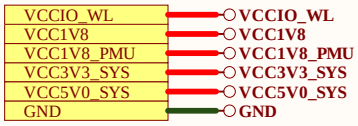
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File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 25/10/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 23

PMIC RK809 Management



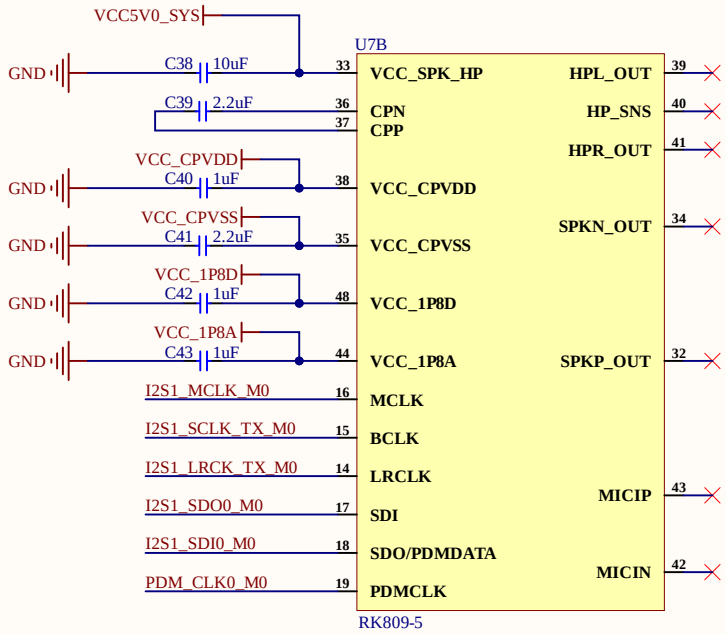


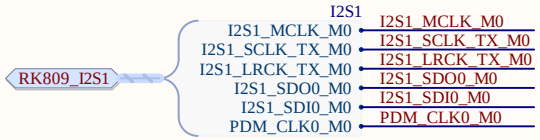




Title: POWER_PMIC_RK809_MGM.SchDoc	
File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 26/11/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 23

PMIC RK809 CODEC



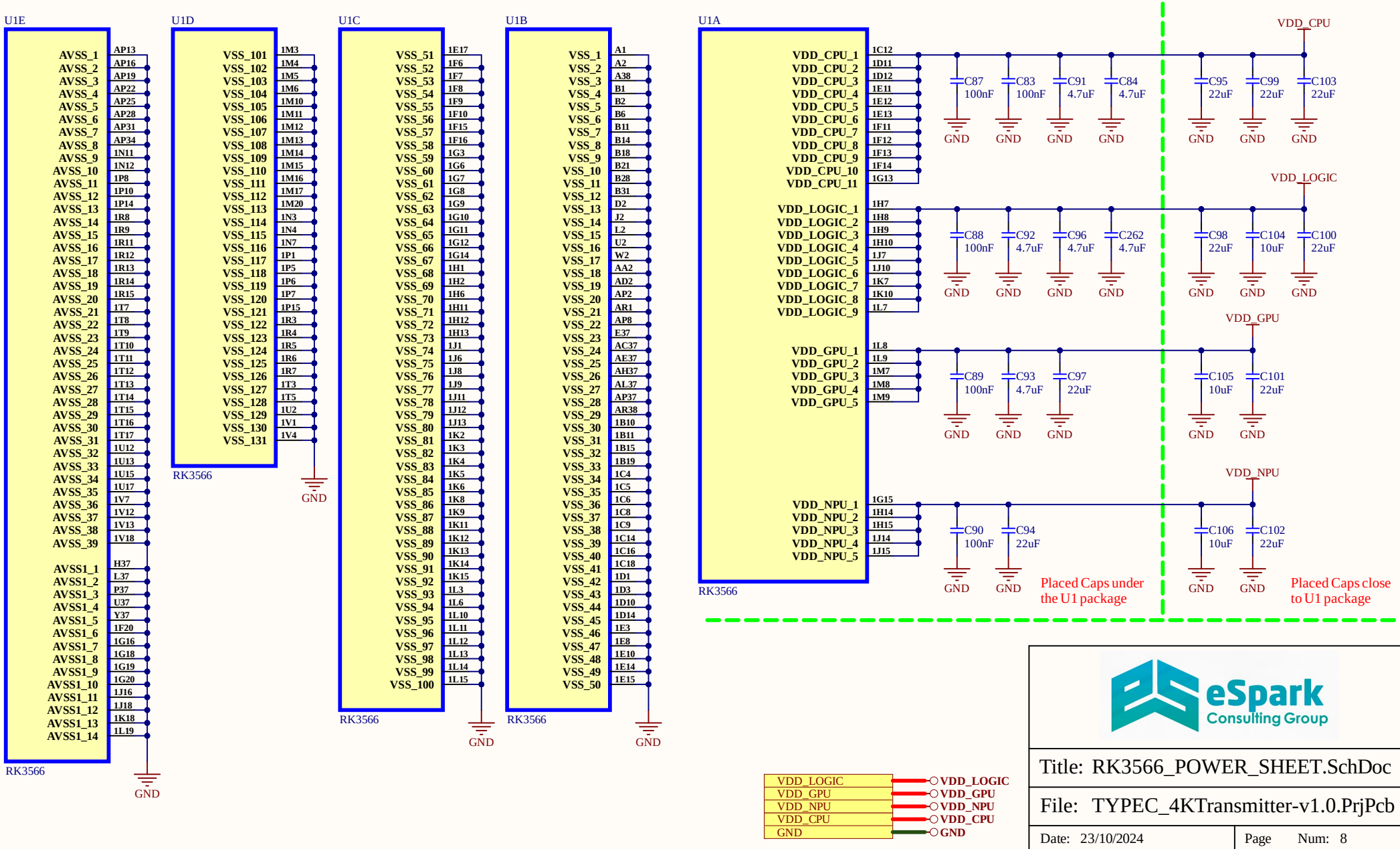




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Title: POWER_PMIC_RK809_CODEC.SchDoc	
File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 25/10/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 23

RK3566_POWER & GROUND



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Title: RK3566_POWER_SHEET.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 23/10/2024	Page Num:	8
Drawn By: Huzaifa Ali	Page Total:	23

RK3566-DDR-LPDDR

U1F

	DDR4	LPDDR4	DDR3	LPDDR3
G2	DDR_DQ0_A	DDR4_DQL0_A	LPDDR4_DQ0_A	DDR3_DQ0	LPDDR3_DQ15
F2	DDR_DQ1_A	DDR4_DQL2_A	LPDDR4_DQ1_A	DDR3_DQ1	LPDDR3_DQ14
F1	DDR_DQ2_A	DDR4_DQL4_A	LPDDR4_DQ2_A	DDR3_DQ2	LPDDR3_DQ10
E1	DDR_DQ3_A	DDR4_DQL6_A	LPDDR4_DQ3_A	DDR3_DQ3	LPDDR3_DQ9
M2	DDR_DQ4_A	DDR4_DQL7_A	LPDDR4_DQ4_A	DDR3_DQ4	LPDDR3_DQ13
K1	DDR_DQ5_A	DDR4_DQL5_A	LPDDR4_DQ5_A	DDR3_DQ5	LPDDR3_DQ12
K2	DDR_DQ6_A	DDR4_DQL3_A	LPDDR4_DQ6_A	DDR3_DQ6	LPDDR3_DQ8
1E1	DDR_DQ7_A	DDR4_DQL1_A	LPDDR4_DQ7_A	DDR3_DQ7	LPDDR3_DQ11
1E2	DDR_DM0_A	DDR4_DML_A	LPDDR4_DM0_A	DDR3_DM0	LPDDR3_DM1
H2	DDR_DQS0P_A	DDR4_DQSL_P_A	LPDDR4_DQS0P_A	DDR3_DQS0P	LPDDR3_DQS1P
H1	DDR_DQS0N_A	DDR4_DQSL_N_A	LPDDR4_DQS0N_A	DDR3_DQS0N	LPDDR3_DQS1N
P2	DDR_DQ8_A	DDR4_DQU3_A	LPDDR4_DQ8_A	DDR3_DQ8	LPDDR3_DQ25
R1	DDR_DQ9_A	DDR4_DQU1_A	LPDDR4_DQ9_A	DDR3_DQ9	LPDDR3_DQ24
R2	DDR_DQ10_A	DDR4_DQU7_A	LPDDR4_DQ10_A	DDR3_DQ10	LPDDR3_DQ28
T2	DDR_DQ11_A	DDR4_DQU5_A	LPDDR4_DQ11_A	DDR3_DQ11	LPDDR3_DQ29
M1	DDR_DQ12_A	DDR4_DQU2_A	LPDDR4_DQ12_A	DDR3_DQ12	LPDDR3_DQ26
1F2	DDR_DQ13_A	DDR4_DQU4_A	LPDDR4_DQ13_A	DDR3_DQ13	LPDDR3_DQ31
1F1	DDR_DQ14_A	DDR4_DQU6_A	LPDDR4_DQ14_A	DDR3_DQ14	LPDDR3_DQ30
1G2	DDR_DQ15_A	DDR4_DQU0_A	LPDDR4_DQ15_A	DDR3_DQ15	LPDDR3_DQ27
1G1	DDR_DM1_A	DDR4_DMU_A	LPDDR4_DM1_A	DDR3_DM1	LPDDR3_DM3
N1	DDR_DQS1P_A	DDR4_DQSU_P_A	LPDDR4_DQS1P_A	DDR3_DQS1P	LPDDR3_DQS3P
N2	DDR_DQS1N_A	DDR4_DQSU_N_A	LPDDR4_DQS1N_A	DDR3_DQS1N	LPDDR3_DQS3N
B12	DDR_DQ0_B	DDR4_DQU7_B	LPDDR4_DQ0_B	DDR3_DQ16	LPDDR3_DQ1
B10	DDR_DQ1_B	DDR4_DQU5_B	LPDDR4_DQ1_B	DDR3_DQ17	LPDDR3_DQ5
1A8	DDR_DQ2_B	DDR4_DQU3_B	LPDDR4_DQ2_B	DDR3_DQ18	LPDDR3_DQ6
1A7	DDR_DQ3_B	DDR4_DQU1_B	LPDDR4_DQ3_B	DDR3_DQ19	LPDDR3_DQ4
B13	DDR_DQ4_B	DDR4_DQU0_B	LPDDR4_DQ4_B	DDR3_DQ20	LPDDR3_DQ2
1A10	DDR_DQ5_B	DDR4_DQU6_B	LPDDR4_DQ5_B	DDR3_DQ21	LPDDR3_DQ3
1B9	DDR_DQ6_B	DDR4_DQU4_B	LPDDR4_DQ6_B	DDR3_DQ22	LPDDR3_DQ7
1A9	DDR_DQ7_B	DDR4_DQU2_B	LPDDR4_DQ7_B	DDR3_DQ23	LPDDR3_DQ0
1B8	DDR_DM0_B	DDR4_DMU_B	LPDDR4_DM0_B	DDR3_DM2	LPDDR3_DM0
A13	DDR_DQS0P_B	DDR4_DQSU_P_B	LPDDR4_DQS0P_B	DDR3_DQS2P	LPDDR3_DQS0P
A12	DDR_DQS0N_B	DDR4_DQSU_N_B	LPDDR4_DQS0N_B	DDR3_DQS2N	LPDDR3_DQS0N
B19	DDR_DQ8_B	DDR4_DQL0_B	LPDDR4_DQ8_B	DDR3_DQ24	LPDDR3_DQ18
A19	DDR_DQ9_B	DDR4_DQL2_B	LPDDR4_DQ9_B	DDR3_DQ25	LPDDR3_DQ19
A20	DDR_DQ10_B	DDR4_DQL4_B	LPDDR4_DQ10_B	DDR3_DQ26	LPDDR3_DQ22
B20	DDR_DQ11_B	DDR4_DQL6_B	LPDDR4_DQ11_B	DDR3_DQ27	LPDDR3_DQ23
A15	DDR_DQ12_B	DDR4_DQL7_B	LPDDR4_DQ12_B	DDR3_DQ28	LPDDR3_DQ16
B15	DDR_DQ13_B	DDR4_DQL5_B	LPDDR4_DQ13_B	DDR3_DQ29	LPDDR3_DQ17
1C10	DDR_DQ14_B	DDR4_DQL1_B	LPDDR4_DQ14_B	DDR3_DQ30	LPDDR3_DQ20
B16	DDR_DQ15_B	DDR4_DQL3_B	LPDDR4_DQ15_B	DDR3_DQ31	LPDDR3_DQ21
1A11	DDR_DM1_B	DDR4_DML_B	LPDDR4_DM1_B	DDR3_DM3	LPDDR3_DM2
A17	DDR_DQS1P_B	DDR4_DQSL_P_B	LPDDR4_DQS1P_B	DDR3_DQS3P	LPDDR3_DQS2P
B17	DDR_DQS1N_B	DDR4_DQSL_N_B	LPDDR4_DQS1N_B	DDR3_DQS3N	LPDDR3_DQS2N

Note:
Except DDR3, other DQ sequences
can not be swap

RK3566

	DDR4	LPDDR4	DDR3	LPDDR3		
A7	DDR4_A0	LPDDR4_CLKP_B	DDR3_A9			LPDDR4_CLK_B_P
D2			DDR3_A2			
C1	DDR4_A1		DDR3_A2			LPDDR4_A1_A
C3	DDR4_A2	LPDDR4_A1_A	DDR3_A4	LPDDR3_A6		LPDDR4_CKE1_A
AC3	DDR4_A3	LPDDR4_CKE1_A	DDR3_A3			
1B6	DDR4_A4	LPDDR4_A3_B	DDR3_BA1	LPDDR3_A3		LPDDR4_A3_B
B8	DDR4_A5	LPDDR4_A5_B	DDR3_A11	LPDDR3_A2		LPDDR4_A5_B
B9	DDR4_A6	LPDDR4_A1_B	DDR3_A13	LPDDR3_A1		LPDDR4_A1_B
A10	DDR4_A7	LPDDR4_ODT0_CA_B	DDR3_A8			LPDDR4_ODT0_CA_B
E2	DDR4_A8	LPDDR4_ODT0_CA_A	DDR3_A6	LPDDR3_A9	AC8	LPDDR4_ODT0_CA_A
B7	DDR4_A9	LPDDR4_CLKN_B	DDR3_A5			LPDDR4_CLK_B_N
A3	DDR4_A10	LPDDR4_CKE0_B	DDR3_A10			LPDDR4_CKE0_B
C2	DDR4_A11	LPDDR4_A0_A	DDR3_A7	LPDDR3_A8	AC11	LPDDR4_A0_A
1B3	DDR4_A12	LPDDR4_A3_A	DDR3_BA2			LPDDR4_A3_A
A9	DDR4_A13	LPDDR4_A0_B	DDR3_A14	LPDDR3_A0	AC13	LPDDR4_A0_B
1B1	DDR4_A14_WEn	LPDDR4_A4_A	DDR3_A15	LPDDR3_A5	AC14	LPDDR4_A4_A
1C1	DDR4_A15_CASn	LPDDR4_A2_A	DDR3_A0			LPDDR4_A2_A
1A1	DDR4_A16_RASn	LPDDR4_A5_A	DDR3_RASn	LPDDR3_A7	AC16	LPDDR4_A5_A
1A2	DDR4_ACTn	LPDDR4_CKE1_B	DDR3_CASn			LPDDR4_CKE1_B
B7	DDR4_BA0	LPDDR4_A2_B	DDR3_A1			LPDDR4_A2_B
1A6	DDR4_BA1	LPDDR4_A4_B	DDR3_A12	LPDDR3_A4	AC19	LPDDR4_A4_B
1A4						
A1	DDR4_BG0	LPDDR4_ODT1_CA_B	DDR3_WEn			
C2	DDR4_BG1	LPDDR4_ODT1_CA_A	DDR3_BA0			
B4	DDR4_CKE	LPDDR4_CKE0_A	DDR3_CKE	LPDDR3_CKE	AC22	LPDDR4_CKE0_A
A5	DDR4_CLKP	LPDDR4_CLKP_A	DDR3_CLKP	LPDDR3_CLKP	AC23	LPDDR4_CLK_A_P
B5	DDR4_CLKN	LPDDR4_CLKN_A	DDR3_CLKN	LPDDR3_CLKN	AC24	LPDDR4_CLK_A_N
B3	DDR4_CS1n	LPDDR4_CS0n_A	DDR3_ODT1	LPDDR3_ODT0	AC25	LPDDR4_CS0n_A
A3	DDR4_CS1n	LPDDR4_CS1n_A	DDR3_CS1n	LPDDR3_ODT1	AC26	LPDDR4_CS1n_A
1A5	DDR4_ODT0	LPDDR4_CS1n_B	DDR3_ODT0	LPDDR3_CS1n	AC27	LPDDR4_CS1n_B
1B5	DDR4_ODT1	LPDDR4_CS0n_B	DDR3_CS0n	LPDDR3_CS0n	AC28	LPDDR4_CS0n_B
1C7	DDR4_RESETn	LPDDR4_RESETn	DDR3_RESETn		AC29	LPDDR4_RESETn

Note: Sequences can not be swap.

	VDDQ	VDDQL
DDR3L	1.35V	1.35V
DDR3	1.50V	1.50V
DDR4	1.20V	1.20V
LPDDR3	1.20V	1.20V
LPDDR4	1.10V	1.10V
LPDDR4x	1.10V	**0.60V**

DDR_RZQ — 1F3 — DDR_RZQ R40 120R VCC_DDR

DDR_VREFOUT — 1H3

DDR_VDDQ_1 — 1D5
DDR_VDDQ_2 — 1D6
DDR_VDDQ_3 — 1D7
DDR_VDDQ_4 — 1D8
DDR_VDDQ_5 — 1D9
DDR_VDDQ_6 — 1E4
DDR_VDDQ_7 — 1F4
DDR_VDDQ_8 — 1G4

VCC_DDR
C108 100nF, C110 100nF, C112 100nF, C114 4.7uF, C116 10uF
GND GND GND GND GND

DDR_VDDQL_1 — 1E6
DDR_VDDQL_2 — 1E7
DDR_VDDQL_3 — 1E9
DDR_VDDQL_4 — 1E5
DDR_VDDQL_5 — 1G5

VCC0V6_DDR
C109 100nF, C111 100nF, C113 100nF, C115 4.7uF, C117 4.7uF
GND GND GND GND GND

Caps should be placed under the U1 package

DDR_AVSS — 1D4 — GND

DM(1..0)A
LPDDR4_DM0_A, LPDDR4_DM1_A → DM(1..0)A
Z50

DM(1..0)B
LPDDR4_DM0_B, LPDDR4_DM1_B → DM(1..0)B
Z50

DQ(15..0)A
LPDDR4_DQ0_A … LPDDR4_DQ15_A → DQ(15..0)A
Z50

DQ(15..0)B
LPDDR4_DQ0_B … LPDDR4_DQ15_B → DQ(15..0)B
Z50

DQS(1..0)A
LPDDR4_DQS0_A_P, LPDDR4_DQS0_A_N, LPDDR4_DQS1_A_P, LPDDR4_DQS1_A_N → DQS(1..0)A

DQS(1..0)B
LPDDR4_DQS0_B_P, LPDDR4_DQS0_B_N, LPDDR4_DQS1_B_P, LPDDR4_DQS1_B_N → DQS(1..0)B

CLK_A
LPDDR4_CLK_A_P, LPDDR4_CLK_A_N → CLK_A

CLK_B
LPDDR4_CLK_B_P, LPDDR4_CLK_B_N → CLK_B

LPDDR4_A
LPDDR4_CKE0_A, LPDDR4_CKE1_A, LPDDR4_ODT0_CA_A, LPDDR4_CS0n_A, LPDDR4_CS1n_A → LPDDR4_A
Z50

LPDDR4_B
LPDDR4_CKE0_B, LPDDR4_CKE1_B, LPDDR4_ODT0_CA_B, LPDDR4_CS0n_B, LPDDR4_CS1n_B → LPDDR4_B
Z50

A(5..0)
LPDDR4_A0_A, LPDDR4_A1_A, LPDDR4_A2_A, LPDDR4_A3_A, LPDDR4_A4_A, LPDDR4_A5_A → A(5..0)
Z50

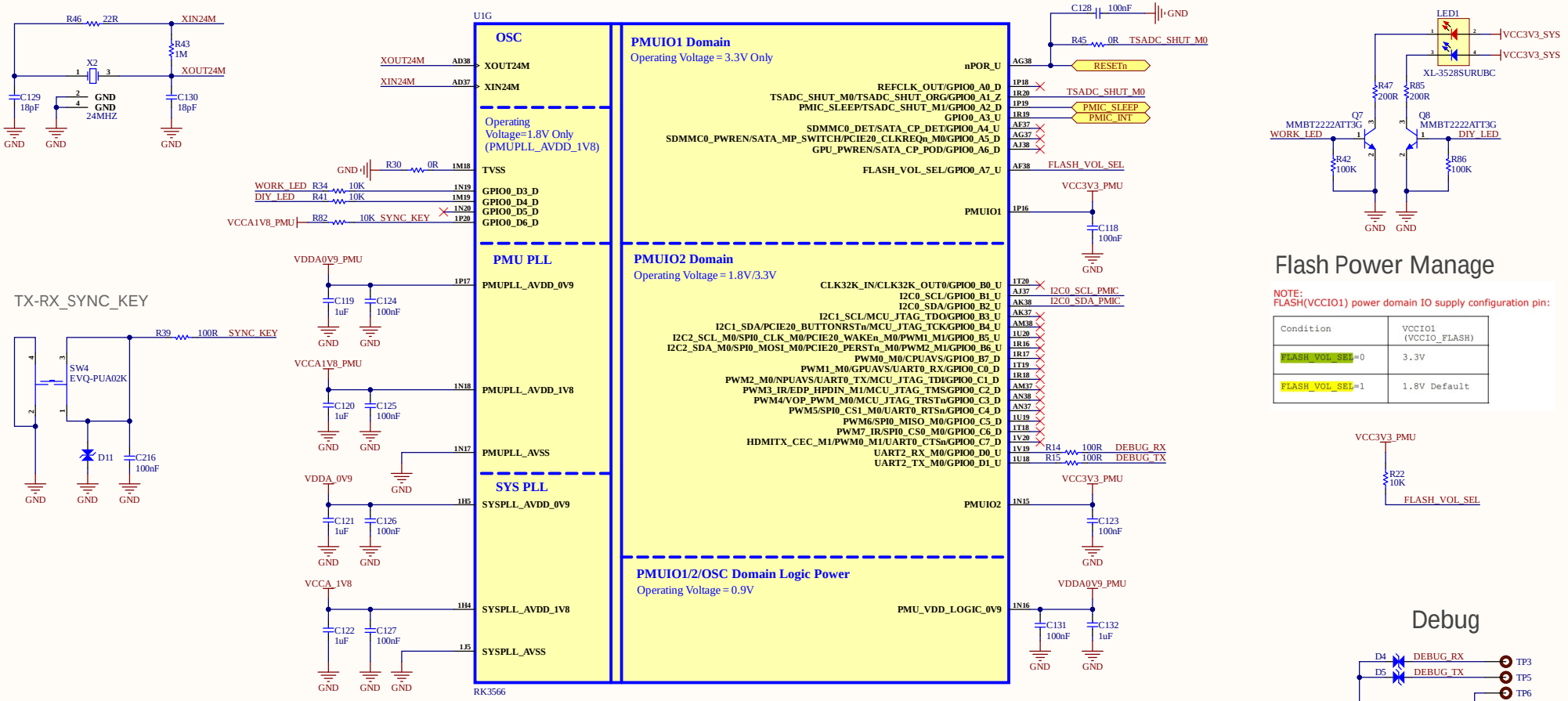
B(5..0)
LPDDR4_A0_B, LPDDR4_A1_B, LPDDR4_A2_B, LPDDR4_A3_B, LPDDR4_A4_B, LPDDR4_A5_B → B(5..0)
Z50

VCC0V6_DDR → VCC0V6_DDR
VCC_DDR → VCC_DDR
GND → GND

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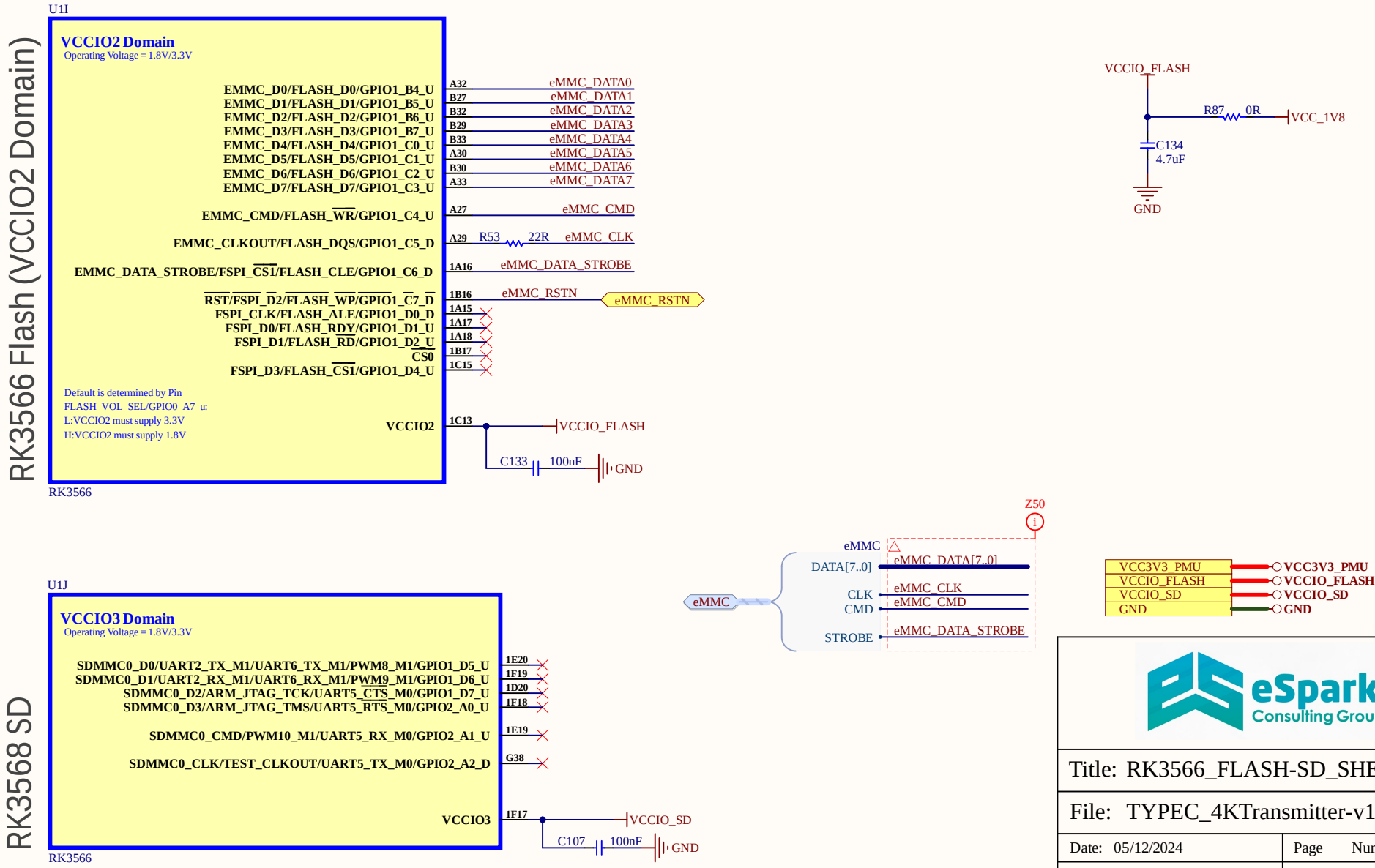
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File:	TYPEC_4KTransmitter-v1.0.PrjPcb
Date: 25/10/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 23

RK3566_OSC/PLL/PMUIO



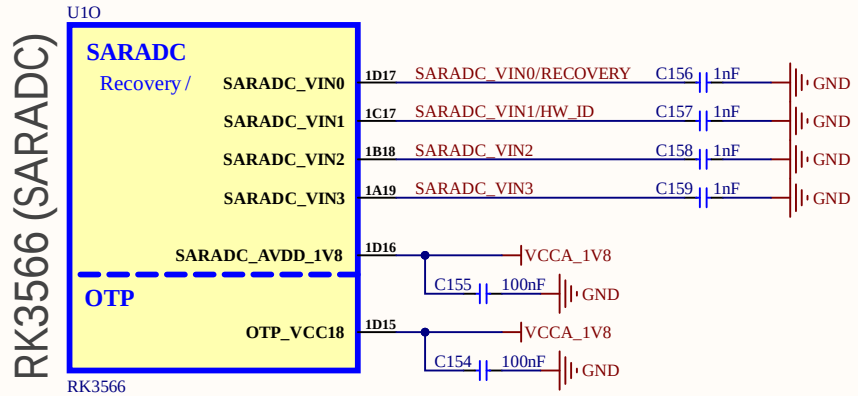
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File:	TYPEC_4KTransmitter-v1.0.PrjPcb
Date: 26/11/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 23

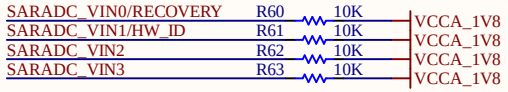
RK3566-eMMC_SPI-FLASH

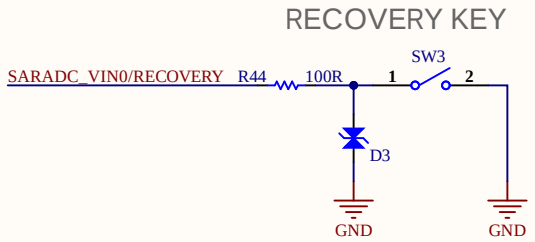


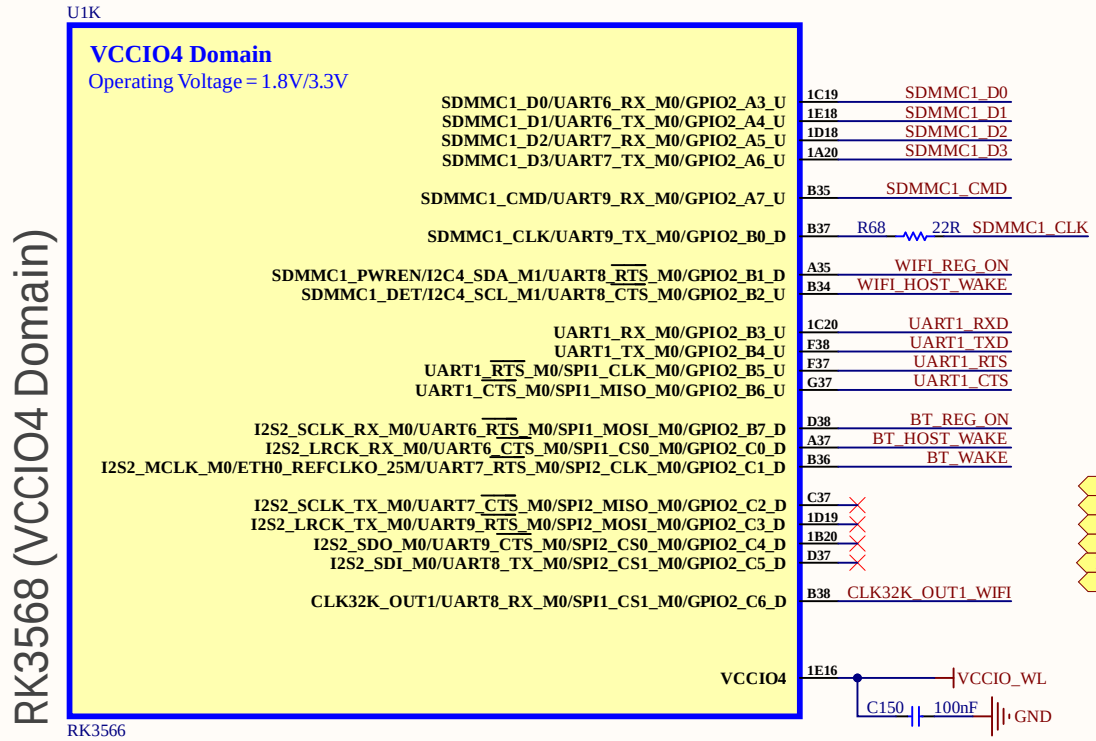
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File:	TYPEC_4KTransmitter-v1.0.PrjPcb
Date: 05/12/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 23

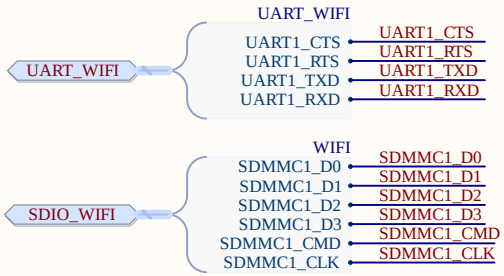
RK3566_SARADC_GPIO



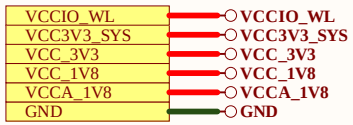












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Title: RK3566_SARADC_GPIO.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 22/10/2024	Page Num:	12
Drawn By: Huzaifa Ali	Page Total:	23

RK3566_USB_PCIe_SHEET

U1U

RK3568 USB3.0/SATAPCIe2.0

USB2.0 OTG_0
HS/FS/LS
(Download Port)

USB_OTG0_DP
USB_OTG0_DM
USB_OTG0_VBUSDET
USB_OTG0_ID

R37 USB_OTG0_D_P
R38 USB_OTG0_D_N
T38 OTG0_VBUSDET
T37
R16 15K GND
R17 10K VCC5V0_SYS
C141 100nF GND

USB2.0 HOST_1
HS/FS/LS
(DP/DM for USB3.0 HOST)

USB_HOST1_DP
USB_HOST1_DM

1J19
1J20

USB2.0 Power
(for OTG_0 and HOST_1)

USB_AVDD1_0V9
USB_AVDD1_1V8
USB_AVDD1_3V3

1J17 VDDA_0V9
C135 100nF GND
1H16 VCCA_1V8
C136 100nF GND
1H18 VCC_3V3
C137 100nF GND

MULTI PHY U_S
(SS for USB3.0 HOST or SATA3_1)

USB3_HOST1_SSTXP/SATA1_TXP
USB3_HOST1_SSTXN/SATA1_TXN
USB3_HOST1_SSRXP/SATA1_RXP
USB3_HOST1_SSRXN/SATA1_RXN

W37
W38
V38
V37

MULTI PHY P_S
(PCIE2.0 or SATA3_2)

PCIE20_TXP/SATA2_TXP
PCIE20_TXN/SATA2_TXN
PCIE20_RXP/SATA2_RXP
PCIE20_RXN/SATA2_RXN
PCIE20_REFCLKP
PCIE20_REFCLKN

AA37
AA38
AB37
AB38
1K19
1K20

MULTI PHY Power
(for MULTI PHY U_S and P_S)

MULTI_PHY_AVDD_0V9
MULTI_PHY_AVDD_1V8

1K16 VDDA_0V9
C138 0.1uF GND
VDDA_0V9 C139 4.7uF GND
1K17 VCCA_1V8
C140 100nF GND
VCCA_1V8 C142 4.7uF GND

RK3568 USB2.0 HOST

U1V

USB2.0
(for HOST_2 and HOST_3)

USB_HOST2_DP
USB_HOST2_DM
USB_HOST3_DP
USB_HOST3_DM
USB_AVDD2_0V9
USB_AVDD2_1V8
USB_AVDD2_3V3

V2 USB2_HOST2_D_P
V1 USB2_HOST2_D_N
Y2
Y1
1J4 VDDA_0V9
C143 100nF GND
1J3 VCCA_1V8
C145 100nF GND
1J2 VCC_3V3
C147 100nF GND

RK3566

USB2.0 USB2_HOST2_D_P USB2_HOST2_D_N
USB2_HOST2_D_P
USB2_HOST2_D_N

OTG0 USB_OTG0 USB_OTG0_D_P USB_OTG0_D_N
USB_OTG0_D_P
USB_OTG0_D_N

VCC5V0_SYS VCC5V0_SYS
VCC_3V3 VCC_3V3
VDDA_0V9 VDDA_0V9
VCCA_1V8 VCCA_1V8
GND GND

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Title: RK3566_USB_PCIe_SHEET.SchDoc	
File: TYPEC_4KTransmitter-v1.0.PrjPcb	
Date: 25/10/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 23

RK3566_VI_INTERFACE

U1M

VCCIO6 Domain
Operating Voltage = 1.8V/3.3V

RK3568 (VCCIO6 Domain)

CIF_D0/EBC_SDDO0/SDMMC2_D0_M0/I2S1_MCLK_M1/VOP_BT656_D0_M1/GPIO3_C6_D	1T2
CIF_D1/EBC_SDDO1/SDMMC2_D1_M0/I2S1_SCLK_TX_M1/VOP_BT656_D1_M1/GPIO3_C7_D	1U1
CIF_D2/EBC_SDDO2/SDMMC2_D2_M0/I2S1_LRCK_TX_M1/VOP_BT656_D2_M1/GPIO3_D0_D	AP1
CIF_D3/EBC_SDDO3/SDMMC2_D3_M0/I2S1_SDO0_M1/VOP_BT656_D3_M1/GPIO3_D1_D	AN2
CIF_D4/EBC_SDDO4/SDMMC2_CMD_M0/I2S1_SDI0_M1/VOP_BT656_D4_M1/GPIO3_D2_D	AM2
CIF_D5/EBC_SDDO5/SDMMC2_CLK_M0/I2S1_SDI1_M1/VOP_BT656_D5_M1/GPIO3_D3_D	AM1
CIF_D6/EBC_SDDO6/SDMMC2_DET_M0/I2S1_SDI2_M1/VOP_BT656_D6_M1/GPIO3_D4_D	1R2
CIF_D7/EBC_SDDO7/SDMMC2_PWREN_M0/I2S1_SDI3_M1/VOP_BT656_D7_M1/GPIO3_D5_D	1T1
CIF_D8/EBC_SDDO8/GMAC1_TXD2_M1/UART1_TX_M1/PDM_CLK0_M1/GPIO3_D6_D	AL2
CIF_D9/EBC_SDDO9/GMAC1_TXD3_M1/UART1_RX_M1/PDM_SDI0_M1/GPIO3_D7_D	AL1
CIF_D10/EBC_SDDO10/GMAC1_TXCLK_M1/PDM_CLK1_M1/GPIO4_A0_D	AK2
CIF_D11/EBC_SDDO11/GMAC1_RXD2_M1/PDM_SDI1_M1/GPIO4_A1_D	1P2
CIF_D12/EBC_SDDO12/GMAC1_RXD3_M1/UART7_TX_M2/PDM_SDI2_M1/GPIO4_A2_D	1R1
CIF_D13/EBC_SDDO13/GMAC1_RXCLK_M1/UART7_RX_M2/PDM_SDI3_M1/GPIO4_A3_D	AJ2
CIF_D14/EBC_SDDO14/GMAC1_TXD0_M1/UART9_TX_M2/I2S2_LRCK_TX_M1/GPIO4_A4_D	1N2
CIF_D15/EBC_SDDO15/GMAC1_TXD1_M1/UART9_RX_M2/I2S2_LRCK_RX_M1/GPIO4_A5_D	AJ1
ISP_FLASHTRIGOUT/EBC_SDCE0/GMAC1_TXEN_M1/SPI3_CS0_M0/I2S1_SCLK_RX_M1/GPIO4_A6_D	AH2
CAM_CLKOUT0/EBC_SDCE1/GMAC1_RXD0_M1/SPI3_CS1_M0/I2S1_LRCK_RX_M1/GPIO4_A7_D	1N1
CAM_CLKOUT1/EBC_SDCE2/GMAC1_RXD1_M1/SPI3_MISO_M0/I2S1_SDO1_M1/GPIO4_B0_D	1L2 — LONT_RESET
ISP_PRELIGHT_TRIG/EBC_SDCE3/GMAC1_RXDV_CRS_M1/I2S1_SDO2_M1/GPIO4_B1_D	AG2
I2C4_SDA_M0/EBC_VCOM/GMAC1_RXER_M1/SPI3_MOSI_M0/I2S2_SDI_M1/GPIO4_B2_D	AF2 — I2C4_SDA_M1
I2C4_SCL_M0/EBC_GDOE/ETH1_REFCLKO_25M_M1/SPI3_CLK_M0/I2S2_SDO_M1/GPIO4_B3_D	AE2 — I2C4_SCL_M1
I2C2_SDA_M1/EBC_GDSP/ISP_FLASH_TRIGIN/VOP_BT656_CLK_M1/GPIO4_B4_D	AF1
I2C2_SCL_M1/EBC_SDSHR/I2S1_SDO3_M1/GPIO4_B5_D	AD1
CIF_HREF/EBC_SDLE/GMAC1_MDC_M1/UART1_RTS_M1/I2S2_MCLK_M1/GPIO4_B6_D	1L1
CIF_VSYNC/EBC_SDOE/GMAC1_MDIO_M1/I2S2_SCLK_TX_M1/GPIO4_B7_D	AC2
CIF_CLKOUT/EBC_GDCLK/PWM11_IR_M1/GPIO4_C0_D	AB2
CIF_CLKIN/EBC_SDCLK/GMAC1_MCLKINOUT_M1/UART1_CTS_M1/I2S2_SCLK_RX_M1/GPIO4_C1_D	AB1
VCCIO6_1	1L4
VCCIO6_2	1L5

RK3566

VCCIO6 — 1L4 / 1L5 — R93 0R — VCC_1V8

C144 100nF — GND

U1P

MIPI CSI RX

RK3568 (MIPI CSI RX)

MIPI_CSI_RX_D0P	AP18 MIPI_CSI_RX_D0_P
MIPI_CSI_RX_D0N	AR18 MIPI_CSI_RX_D0_N
MIPI_CSI_RX_D1P	AR17 MIPI_CSI_RX_D1_P
MIPI_CSI_RX_D1N	AP17 MIPI_CSI_RX_D1_N
MIPI_CSI_RX_D2P	AP15 MIPI_CSI_RX_D2_P
MIPI_CSI_RX_D2N	AR15 MIPI_CSI_RX_D2_N
MIPI_CSI_RX_D3P	AR14 MIPI_CSI_RX_D3_P
MIPI_CSI_RX_D3N	AP14 MIPI_CSI_RX_D3_N
MIPI_CSI_RX_CLK0P	1V9 MIPI_CSI_RX_CLK0_P
MIPI_CSI_RX_CLK0N	1U9 MIPI_CSI_RX_CLK0_N
MIPI_CSI_RX_CLK1P	1V8
MIPI_CSI_RX_CLK1N	1U8
MIPI_CSI_RX_AVDD_0V9	1N9 — VDDA0V9_IMAGE
MIPI_CSI_RX_AVDD_1V8	1P9 — VCCA1V8_IMAGE

RK3566

MIPI_CSI_RX_AVDD_0V9: C148 0.1uF — GND; C149 1uF — GND

MIPI_CSI_RX_AVDD_1V8: C168 100nF — GND; C169 1uF — GND

VCC_1V8 — R111 2.2K — I2C4_SCL_M1

VCC_1V8 — R112 2.2K — I2C4_SDA_M1

I2C4 — I2C4_SDA_M1 — I2C4_SCL_M1

MIPI_CSI

MIPI_CSI_RX_D0_P	MIPI_CSI_RX_D0_P
MIPI_CSI_RX_D0_N	MIPI_CSI_RX_D0_N
MIPI_CSI_RX_D1_P	MIPI_CSI_RX_D1_P
MIPI_CSI_RX_D1_N	MIPI_CSI_RX_D1_N
MIPI_CSI_RX_D2_P	MIPI_CSI_RX_D2_P
MIPI_CSI_RX_D2_N	MIPI_CSI_RX_D2_N
MIPI_CSI_RX_D3_P	MIPI_CSI_RX_D3_P
MIPI_CSI_RX_D3_N	MIPI_CSI_RX_D3_N
MIPI_CSI_RX_CLK0_P	MIPI_CSI_RX_CLK0_P
MIPI_CSI_RX_CLK0_N	MIPI_CSI_RX_CLK0_N

VCC_3V3	VCC_3V3
VDDA0V9_IMAGE	VDDA0V9_IMAGE
VCCA1V8_IMAGE	VCCA1V8_IMAGE
VCC_1V8	VCC_1V8
GND	GND

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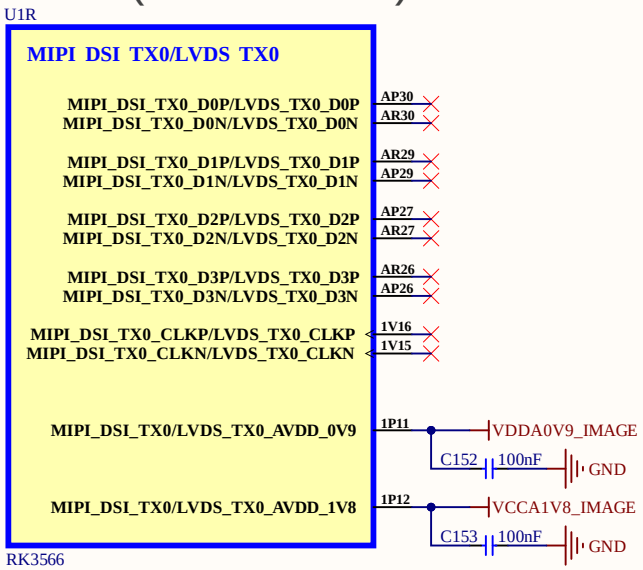
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File: TYPEC_4KTransmitter-v1.0.PrjPcb

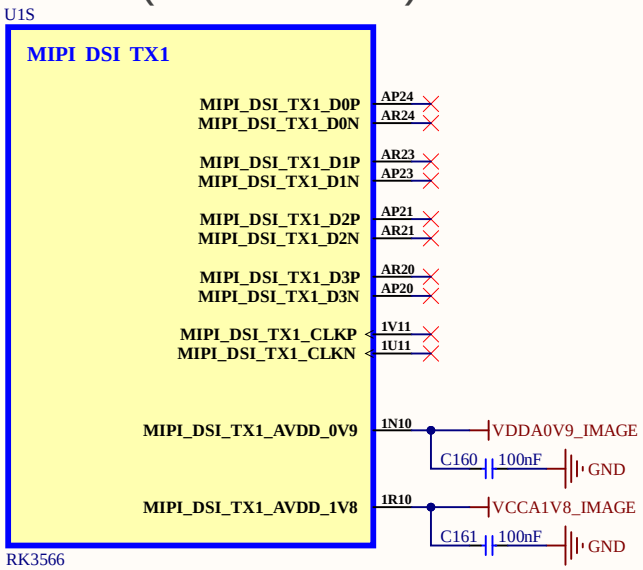
Date: 20/11/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 23

RK3566_VO_INTERFACE_1

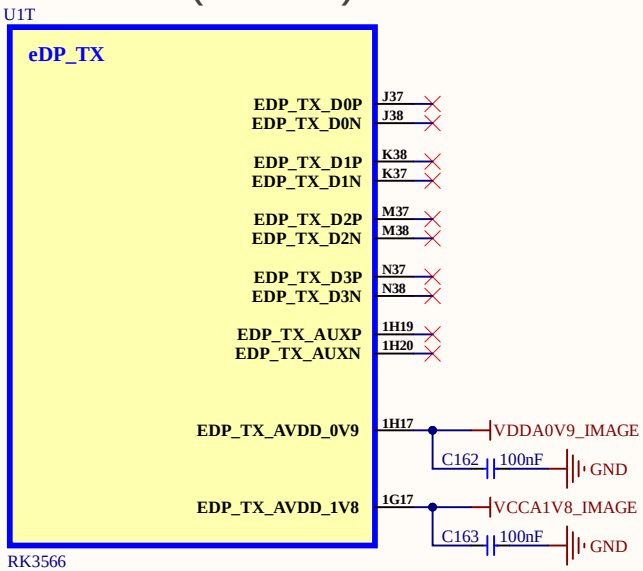
RK3566 (MIPI DSI TX0)



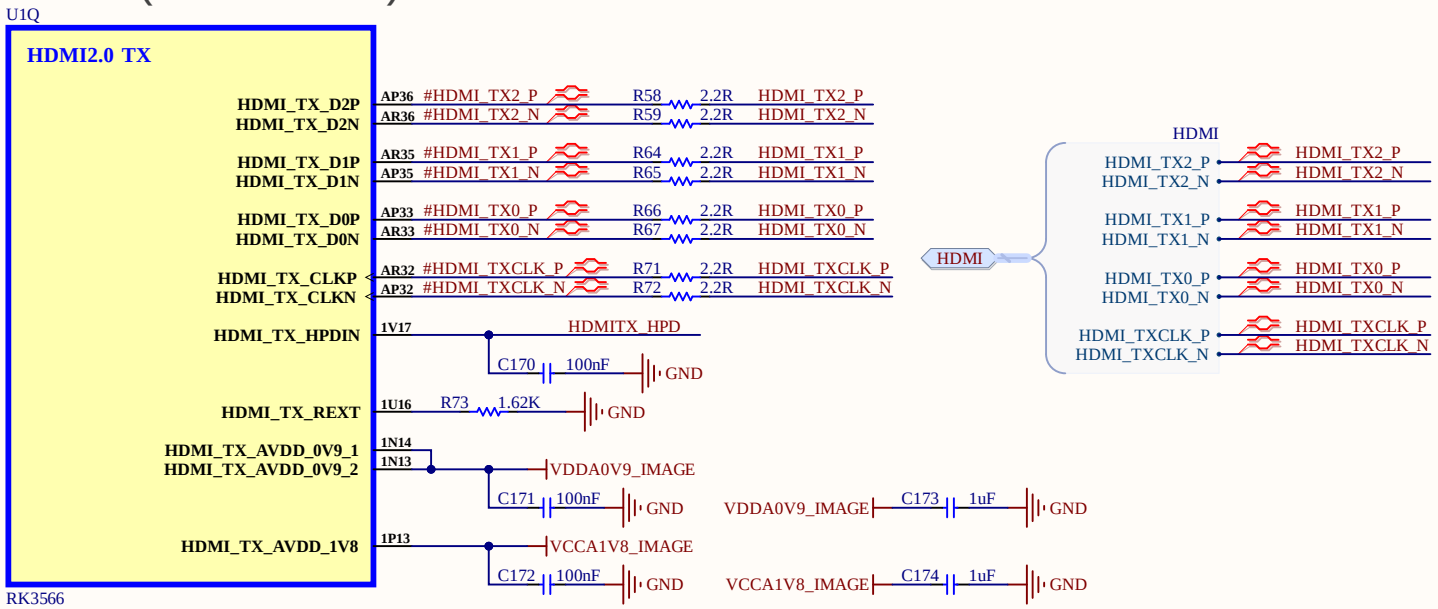
RK3566 (MIPI DSI TX1)

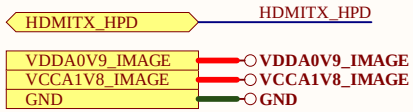


RK3566 (eDP TX)



RK3566 (HDMI2.0 TX)





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Title: RK3566_VO_SHEET_1.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 26/11/2024	Page Num:	15
Drawn By: Huzaifa Ali	Page Total:	23

RK3566_VO_INTERFACE_2

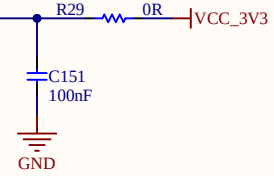
RK3566 (VCCIO5 Domain)

U1L

VCCIO5 Domain
Operating Voltage = 1.8V/3.3V

VOP_BT1120_D0/SPI1_CS0_M1/SDMMC2_D0_M1/GPIO3_A1_D	1U5
VOP_BT1120_D1/GMAC1_TXD2_M0/I2S3_MCLK_M0/SDMMC2_D1_M1/GPIO3_A2_D	AR10
VOP_BT1120_D2/GMAC1_TXD3_M0/I2S3_SCLK_M0/SDMMC2_D2_M1/GPIO3_A3_D	AP10
VOP_BT1120_D3/GMAC1_RXD2_M0/I2S3_LRCK_M0/SDMMC2_D3_M1/GPIO3_A4_D	AR9
VOP_BT1120_D4/GMAC1_RXD3_M0/I2S3_SDO_M0/SDMMC2_CMD_M1/GPIO3_A5_D	AP9
VOP_BT1120_CLK/GMAC1_TXCLK_M0/I2S3_SDI_M0/SDMMC2_CLK_M1/GPIO3_A6_D	1U4
VOP_BT1120_D5/GMAC1_RXCLK_M0/SDMMC2_DET_M1/GPIO3_A7_D	1V3
VOP_BT1120_D6/ETH1_REFCLKO_25M_M0/SDMMC2_PWREN_M1/GPIO3_B0_D	AR7
VOP_BT1120_D7/GMAC1_RXD0_M0/UART4_RX_M1/PWM8_M0/GPIO3_B1_D	AP7
VOP_BT1120_D8/GMAC1_RXD1_M0/UART4_TX_M1/PWM9_M0/GPIO3_B2_D	AR6
VOP_BT1120_D9/GMAC1_RXDV_CRS_M0/I2C5_SCL_M0/PDM_SDI0_M2/GPIO3_B3_D	AP6
VOP_BT1120_D10/GMAC1_RXER_M0/I2C5_SDA_M0/PDM_SDI1_M2/GPIO3_B4_D	1U3
VOP_BT1120_D11/GMAC1_TXD0_M0/I2C3_SCL_M1/PWM10_M0/GPIO3_B5_D	1T4
VOP_BT1120_D12/GMAC1_TXD1_M0/I2C3_SDA_M1/PWM11_IR_M0/GPIO3_B6_D	1V2
PWM12_M0/GMAC1_TXEN_M0/UART3_TX_M1/PDM_SDI2_M2/GPIO3_B7_D	AP5
PWM13_M0/GMAC1_MCLKINOUT_M0/UART3_RX_M1/PDM_SDI3_M2/GPIO3_C0_D	AR4
VOP_BT1120_D13/SPI1_MOSI_M1/PCIE20_PERST_M1/I2S1_SDO2_M2/GPIO3_C1_D	AP4
VOP_BT1120_D14/SPI1_MISO_M1/UART5_TX_M1/I2S1_SDO3_M2/GPIO3_C2_D	AP3
VOP_BT1120_D15/SPI1_CLK_M1/UART5_RX_M1/I2S1_SCLK_RX_M2/GPIO3_C3_D	AR2
PWM14_M0/VOP_PWM_M1/GMAC1_MDC_M0/UART7_TX_M1/PDM_CLK1_M2/GPIO3_C4_D	1P3
PWM15_IR_M0/SPDIF_TX_M1/GMAC1_MDIO_M0/UART7_RX_M1/I2S1_LRCK_RX_M2/GPIO3_C5_D	1P4
VCCIO5_1	1N5
VCCIO5_2	1N6

RK3566





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Title: RK3566_VO_SHEET_2.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 24/10/2024	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 23

RK3566-Audio-Interface

RK3568 (VCCIO1 Domain)

U1H

VCCIO1 Domain
Operating Voltage = 1.8V/3.3V

I2C3_SDA_M0/UART3_RX_M0/AUDIOPWM_LOUT_P/GPIO1_A0_U	A22
I2C3_SCL_M0/UART3_TX_M0/AUDIOPWM_LOUT_N/GPIO1_A1_U	B22
I2S1_MCLK_M0/UART3_RTS_M0/SCR_CLK/GPIO1_A2_D	A23 — I2S1_MCLK_M0
I2S1_SCLK_TX_M0/UART3_CTS_M0/SCR_IO/GPIO1_A3_D	B23 — I2S1_SCLK_TX_M0
I2S1_SCLK_RX_M0/UART4_RX_M0/PDM_CLK1_M0/SPDIF_TX_M0/GPIO1_A4_D	1A13
I2S1_LRCK_TX_M0/UART4_RTS_M0/SCR_RST/GPIO1_A5_D	B24 — I2S1_LRCK_TX_M0
I2S1_LRCK_RX_M0/UART4_TX_M0/PDM_CLK0_M0/AUDIOPWM_ROUT_P/GPIO1_A6_D	1A14 — PDM_CLK0_M0
I2S1_SDO0_M0/UART4_CTS_M0/SCR_DET/AUDIOPWM_ROUT_N/GPIO1_A7_D	B25 — I2S1_SDO0_M0
I2S1_SDO1_M0/I2S1_SDI3_M0/PDM_SDI3_M0/PCIE20_CLKREQ_M2/GPIO1_B0_D	1B13
I2S1_SDO2_M0/I2S1_SDI2_M0/PDM_SDI2_M0/PCIE20_WAKE_M2/GPIO1_B1_D	A26
I2S1_SDO3_M0/I2S1_SDI1_M0/PDM_SDI1_M0/PCIE20_PERST_M2/GPIO1_B2_D	B26
I2S1_SDI0_M0/PDM_SDI0_M0/GPIO1_B3_D	1B14 — I2S1_SDI0_M0
VCCIO1	1D13 — VCCIO_ACODEC; C180 100nF GND

RK3566

RK3568 (VCCIO7 Domain)

U1N

VCCIO7 Domain
Operating Voltage = 1.8V/3.3V

PWM14_M1/SPI3_CLK_M1/I2S3_MCLK_M1/GPIO4_C2_D	1U7 R119 0R I2S3_MCLK_M1
PWM15_IR_M1/SPI3_MOSI_M1/I2S3_SCLK_M1/GPIO4_C3_D	AP12 R120 0R I2S3_SCLK_M1
EDP_HPDIN_M0/SPDIF_TX_M2/SATA2_ACT_LED/I2S3_LRCK_M1/GPIO4_C4_D	1V6 R121 0R I2S3_LRCK_M1
PWM12_M1/SPI3_MISO_M1/SATA1_ACT_LED/UART9_TX_M1/I2S3_SDO_M1/GPIO4_C5_D	1U6
PWM13_M1/SPI3_CS0_M1/SATA0_ACT_LED/UART9_RX_M1/I2S3_SDI_M1/GPIO4_C6_D	1T6 R123 0R I2S3_SDO_M1
HDMITX_SCL/I2C5_SCL_M1/GPIO4_C7_U	AR12 HDMITX_SCL
HDMITX_SDA/I2C5_SDA_M1/GPIO4_D0_U	AP11 HDMITX_SDA
HDMITX_CEC_M0/SPI3_CS1_M1/GPIO4_D1_U	1V5 HDMITX_CEC
VCCIO7	1N8 — VCC_3V3; C181 100nF GND

RK3566

I2S3
LONT_I2S3 — I2S3_MCLK_M1, I2S3_SCLK_M1, I2S3_LRCK_M1, I2S3_SDO_M1

I2C5
HDMI_I2C5 — HDMITX_SCL, HDMITX_SDA

I2S1
RK809_I2S1 — I2S1_MCLK_M0, I2S1_SCLK_TX_M0, I2S1_LRCK_TX_M0, I2S1_SDO0_M0, I2S1_SDI0_M0, PDM_CLK0_M0

HDMITX_CEC — HDMITX_CEC

VCCIO_ACODEC — VCCIO_ACODEC
VCC_3V3 — VCC_3V3
GND — GND

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Title: RK3566_AUDIO_SHEET.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 21/11/2024	Page Num: 17	
Drawn By: Huzaifa Ali	Page Total: 23	

DRAM_LPDDR4/LPDDR4X_SHEET

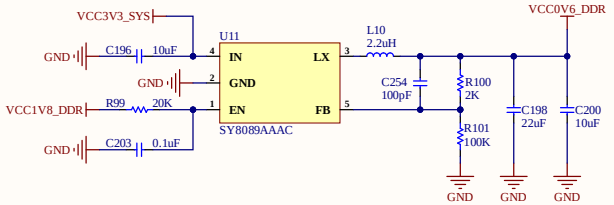
512M32 (16Gb)

16Gb = 64Mb x16DQ x2CHANNEL

PCB Layout Length Matching Guideline

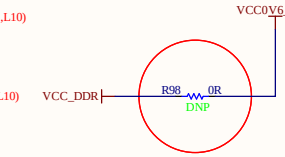
LPDDR4-4000 Length Matching Guideline

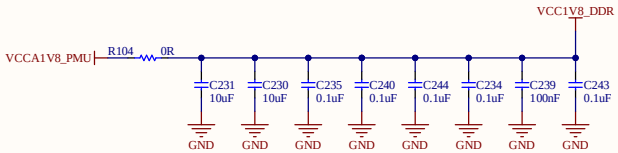
LPDDR4 signal (each 16-bit channel)	Group	PCB and package prop delay		Considerations
		Min	Max	
CK_t/CK_c	Clock	Short as possible	200 ps	
CA5, CA4	Address/ Command/ Control	CK_t - 50 ps	CK_t + 50 ps	Match CA5, CA4 within 2 ps
CA3, CA2, CA1, CA0				Match CA3, CA2, CA1, CA0 within 2 ps
CKE1, CKE0, CS1, CS0				Match CKE1, CKE0, CS1, CS0 within 2 ps
DQS0_t/DQS0_c	Byte 0 - DQS	CK_t - 75 ps	CK_t + 75 ps	Match the true/complement signals of DQS within 1 ps.
DM0	Byte 0 - Data	DQS0_t -50 ps	DQS0_t +50 ps	
DQ[7:0]				
DQS1_t/DQS1_c	Byte 1 - DQS	CK_t - 75 ps	CK_t + 75 ps	
DM1	Byte 1 - Data	DQS1_t -50 ps	DQS1_t +50 ps	
DQ[15:8]				

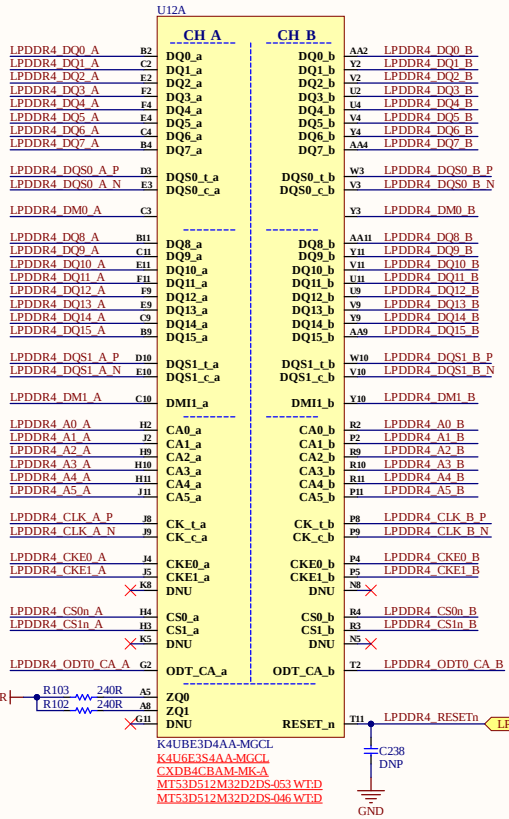


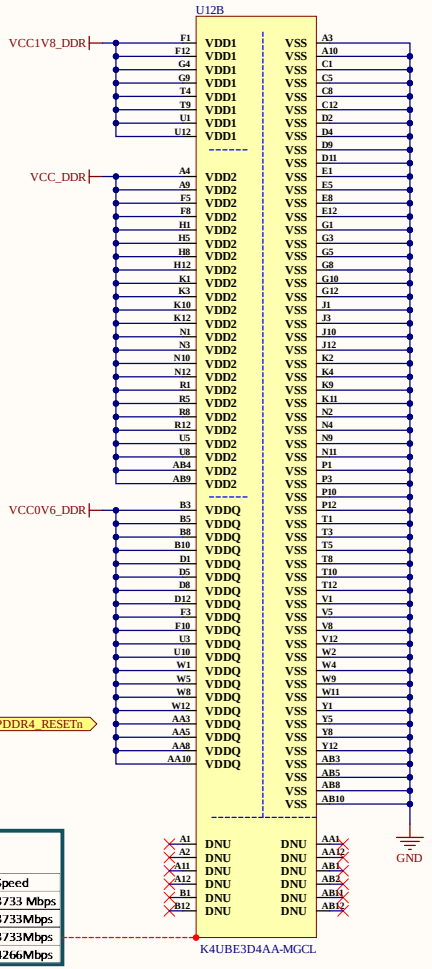
LPDDR4 MOUNT R98 REMOVE (U11,L10)

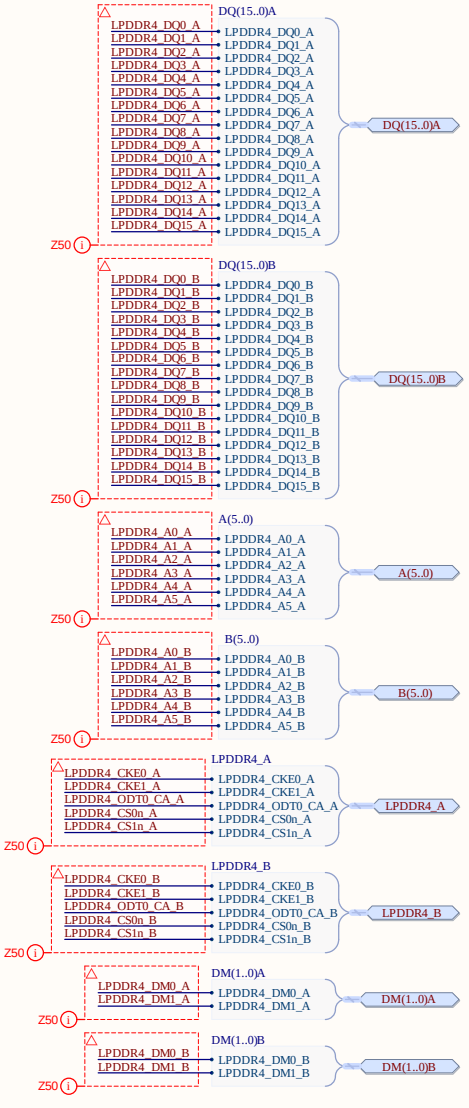
LPDDR4X REMOVE R98 MOUNT (U11,L10)





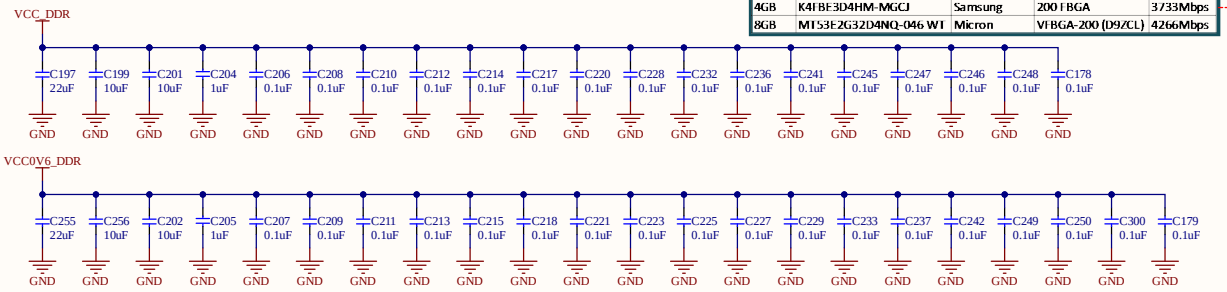




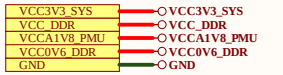


Assembly Variant Part Nubers

Size	MPN	Manufacturer	Package	Speed
1GB	K4F8F304HB-MGCJ	Samsung	200 FBGA	3733 Mbps
2GB	K4F6F3S4HM-MGCJ	Samsung	200 FBGA	3733Mbps
4GB	K4FBE3D4HM-MGCJ	Samsung	200 FBGA	3733Mbps
8GB	MT53E2G32D4NQ-046 WT	Micron	VFBGA-200 (D9ZCL)	4266Mbps



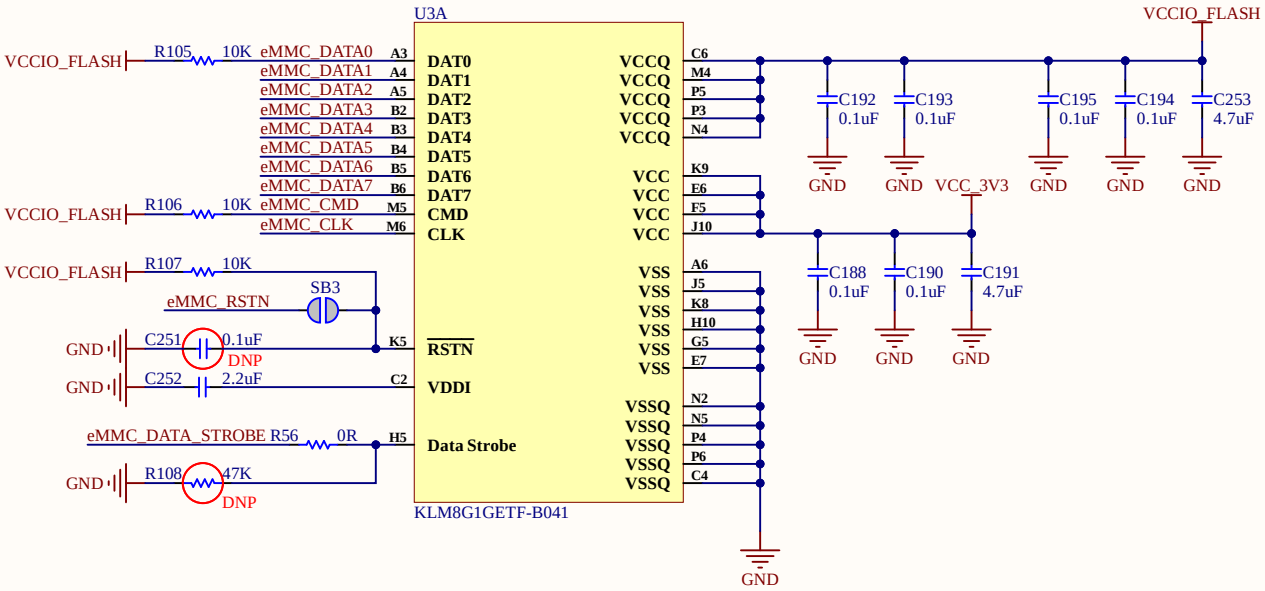


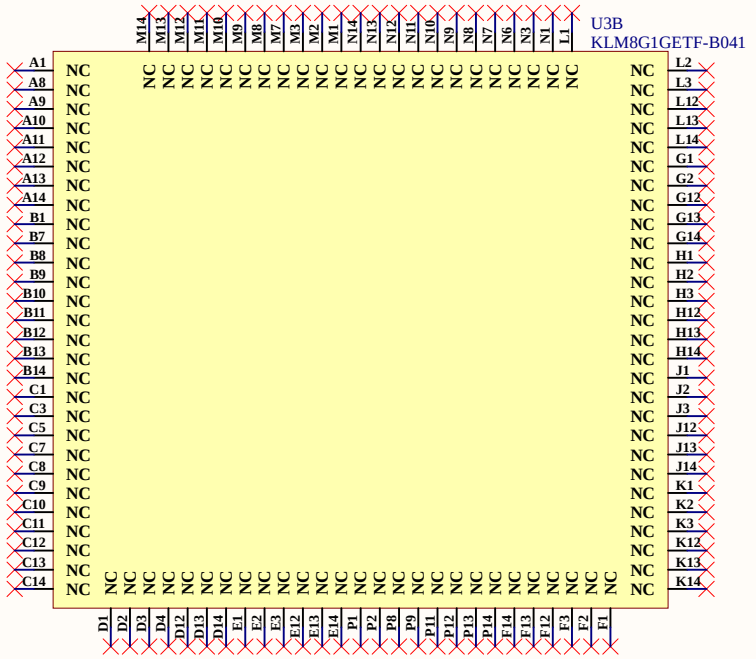


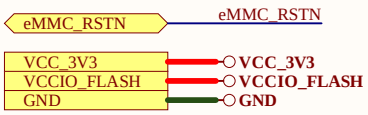
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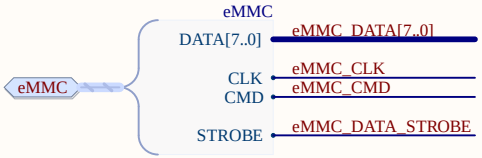
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File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 03/12/2024	Page	Num: 18
Drawn By: Huzaifa Ali	Page	Total: 23

eMMC_SPI-FLASH_SHEET









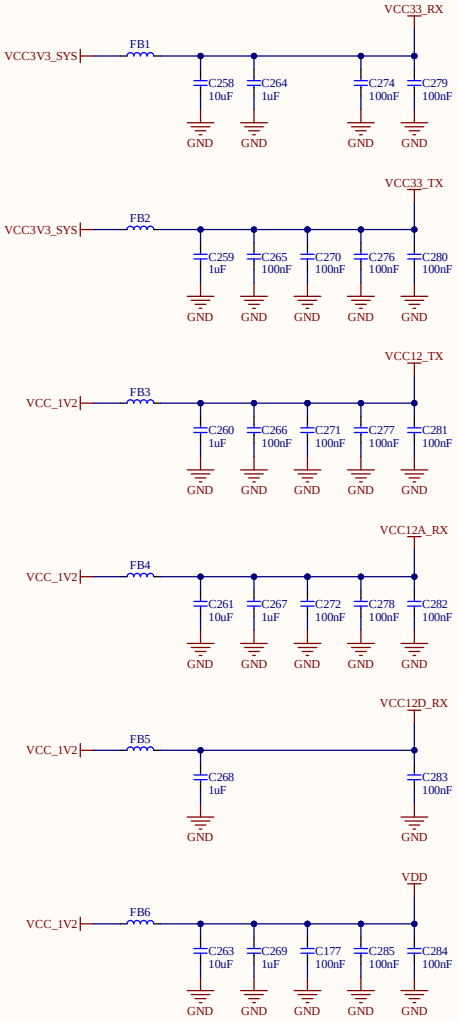
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Date: 29/11/2024	Page Num: 19
Drawn By: Huzaifa Ali	Page Total: 23

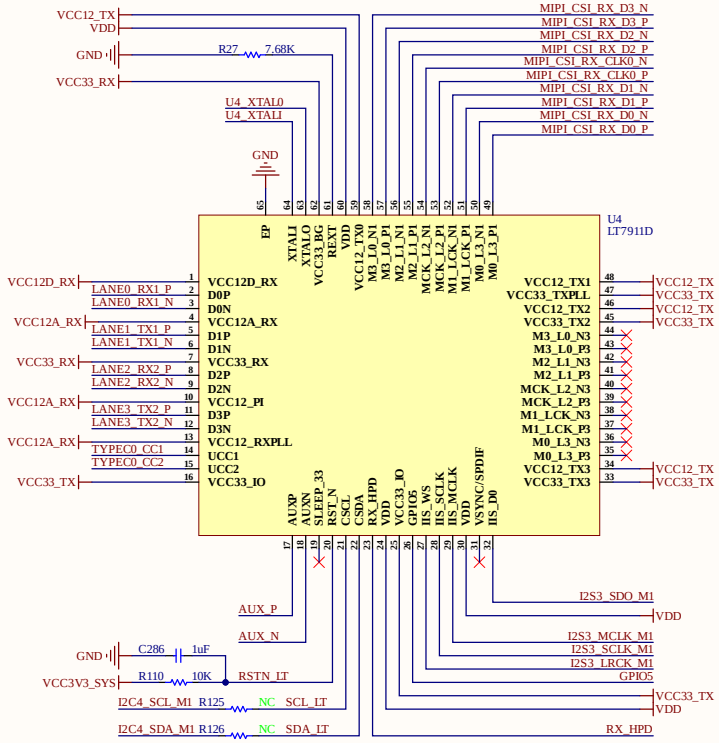
TYPE-C_TO_MIPI-CSI_INTERFACE

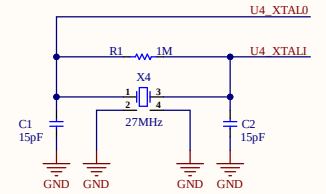
The lane of the 7911D supports swap, so the connection order defined in our DEMO and specification is different.

You can choose one of the following two connection orders, and it is recommended to choose the DEMO line order first for easier firmware creation.

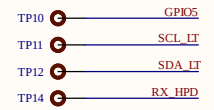
Type C Receptacle	SPEC	DEMO
A2/A3	3P/3N LANE3	1P/1N LANE1
A10/A11	1N/1P LANE1	2P/2N LANE2
B2/B3	0P/0N LANE0	3P/3N LANE3
B10/B11	2N/2P LANE2	0P/0N LANE0

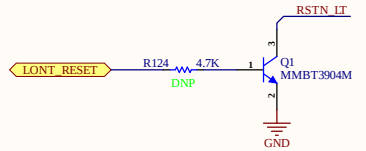


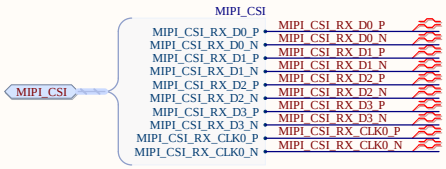


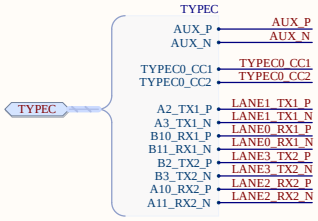


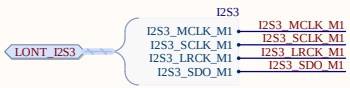
TEST POINTs

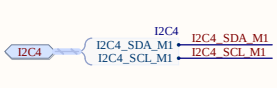










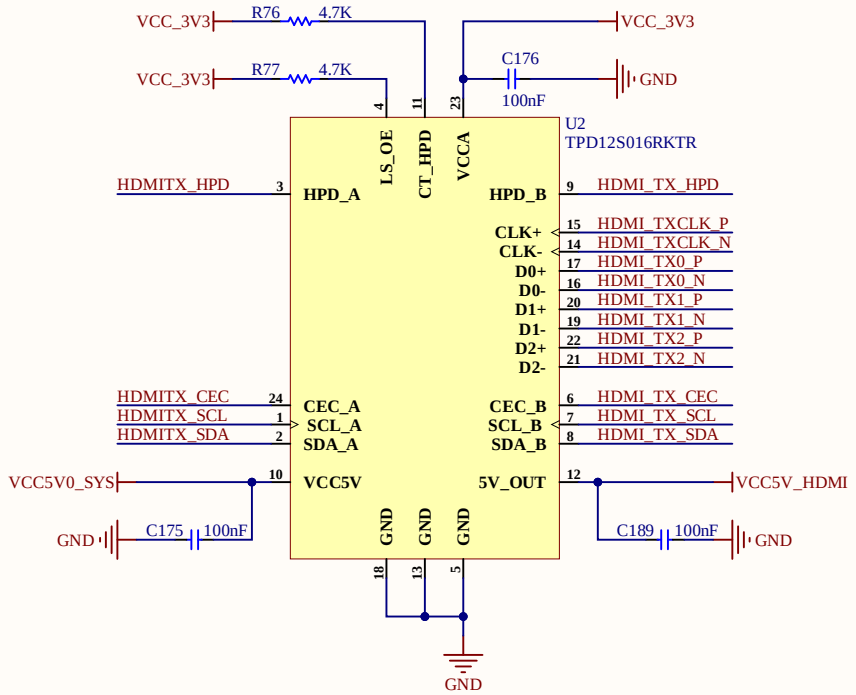


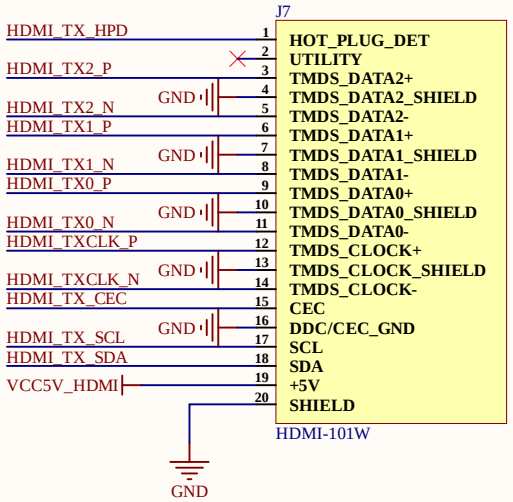


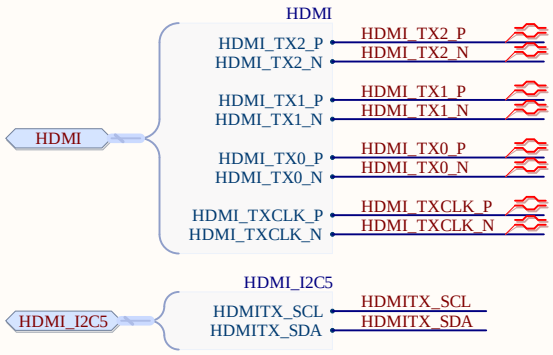
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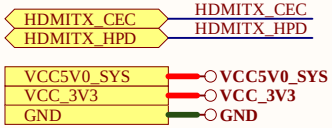
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File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 21/11/2024	Page	Num: 20
Drawn By: Huzaifa Ali	Page	Total: 23

HDMI2.0-INTERFACE





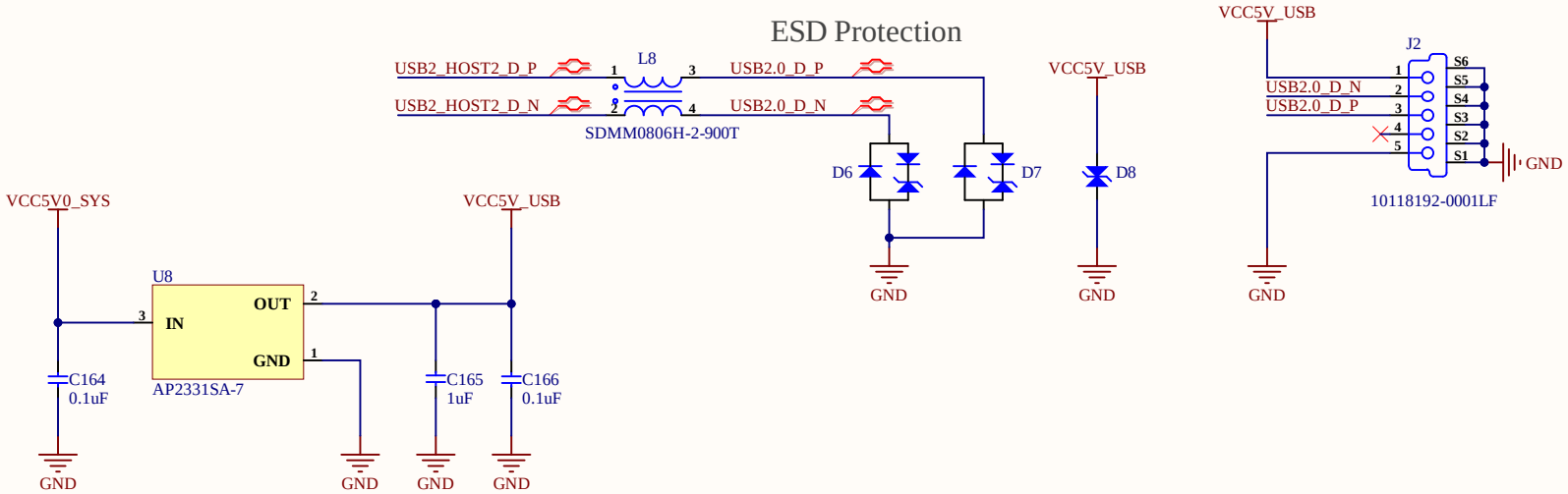


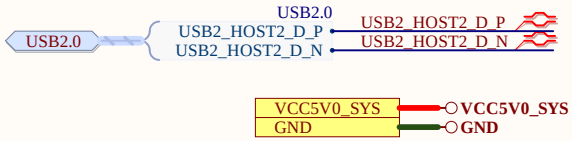


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Title: HDMI2.0_SHEET.SchDoc		
File: TYPEC_4KTransmitter-v1.0.PrjPcb		
Date: 20/11/2024	Page	Num: 21
Drawn By: Huzaifa Ali	Page	Total: 23

USB2.0-INTERFACE

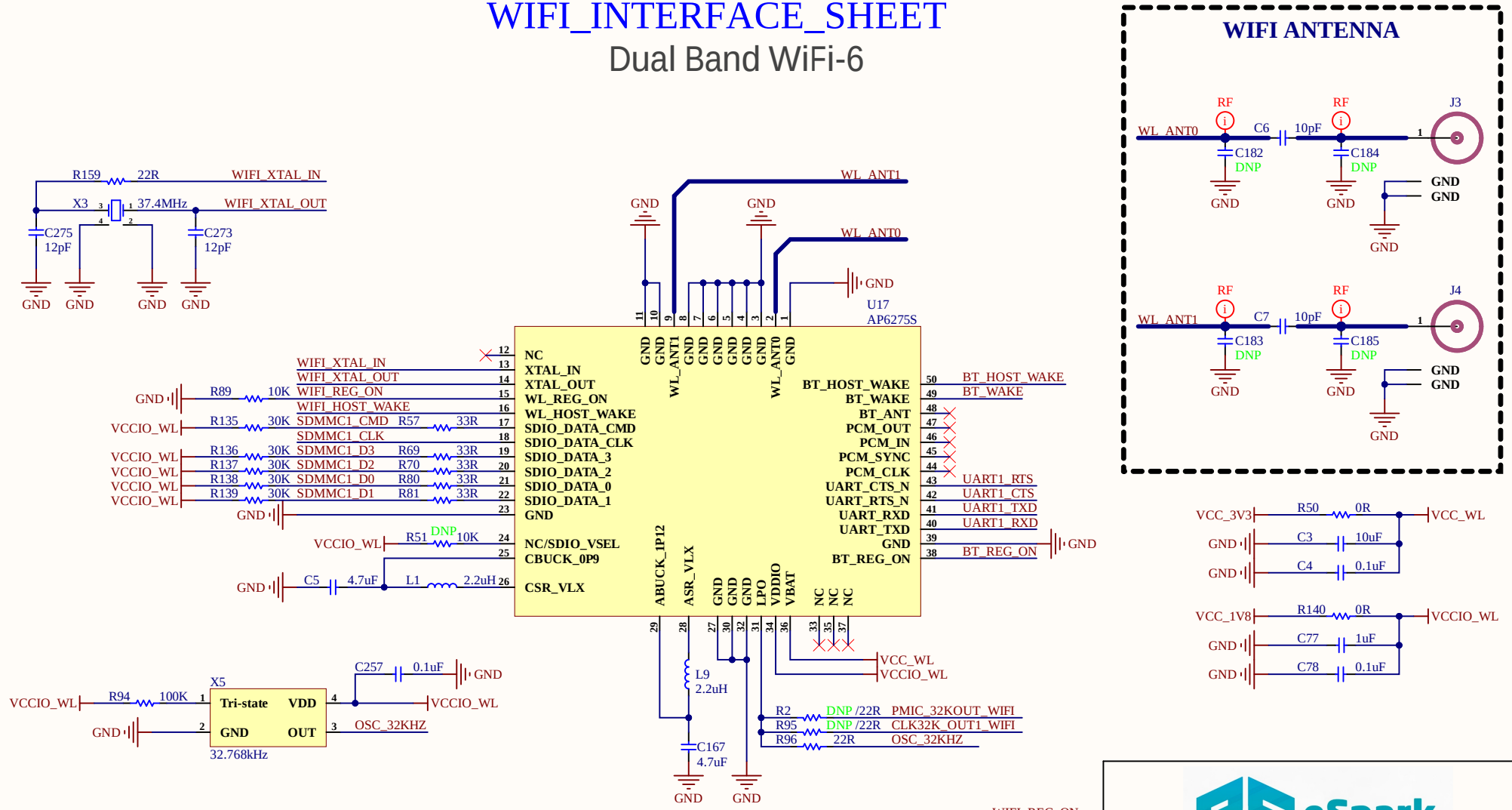




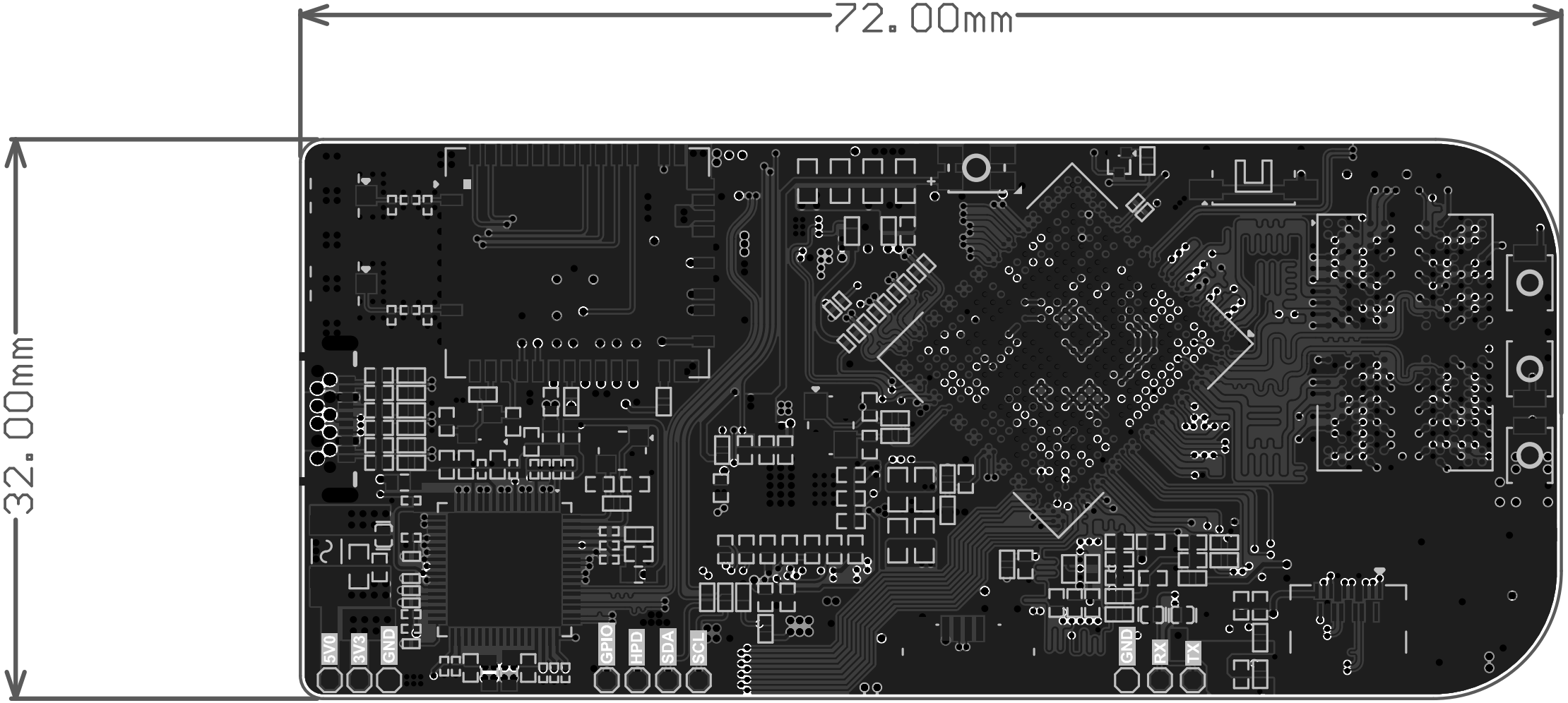
Title: USB2.0_SHEET.SchDoc	
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Date: 05/12/2024	Page Num: 22
Drawn By: Huzaifa Ali	Page Total: 23

WIFI_INTERFACE_SHEET

Dual Band WiFi-6



Title:	WIFI-6_SHEET.SchDoc
File:	TYPEC_4KTransmitter-v1.0.PrjPcb
Date: 06/12/2024	Page Num: 23
Drawn By: Huzaifa Ali	Page Total: 23


72.00mm
32.00mm
5V0
3V3
GND
GPIO
HPD
SDA
SCL
GND
RX
TX